Whitecliffe

IT8105 Machine Learning

Final Report

Scenario-based Evaluative Report of Machine Learning Solution

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Abstract

Recorded crime in the City of Chicago is highly clustered, with the busiest 20% of 2.2 km grid cells accounting for approximately half of all recorded incidents. This study evaluates whether an area-level forecasting approach using a Spatial Graph Convolutional Network (GCN) can achieve more accurate predictions for the two forecasting tasks: (1) dominant crime type during the following 28 days and (2) whether a cell is among the top 20% for recorded victimisation-related incidents, versus persistence, kernel density, logistic regression (LR), SVM, random forest, MLP, and graph-attention baselines. Features use only past periods; models are tuned using validation periods and tested in chronological order using ten seeds, spatial-block bootstrap intervals, and permutation testing. A review found that the label for Phase 1 of the Assessment 1 pilot was present in its features, and LR matched the pilot's Phase 2 result. For one year (228,595 rows, 157 cells), LR achieved an average precision of 0.983 and macro-F1 of 0.951 for cells with high victimisation, exceeding the pre-registered criterion of 0.70, but persistence achieved 0.973 and no learned model performed significantly better than persistence. The GCN reached 0.885 and a residual variant 0.975; neither made use of neighbouring cells productively, and a three-year replication (732,017 rows, six test periods) yielded the same ranking. The prediction of the most prevalent crime was a challenge (best macro-F1 at 0.50). The results using an API prototype are identical to the off-line version. A transparent method, with a human in the loop, is proposed.

Contents

Abstract **i**

List of Abbreviations **viii**

1 Introduction **1**

1.1 Problem Statement 1

1.2 Problem Significance 2

1.3 Research Objectives 3

1.4 Structure of the Report 4

2 Literature Review **5**

2.1 Background 5

2.2 Related Work 5

2.3 Research Gaps and Challenges 7

2.4 Theoretical Framework 8

2.5 Chapter Summary 8

3 Methodology **9**

3.1 Overall Approach 9

3.2 Data Collection and Preprocessing 10

3.2.1 Data Cleaning 11

3.2.2 Data Transformation & Feature Engineering 11

3.2.3 Class Imbalance Resolution 13

3.2.4 Feature Selection 14

3.2.5 Dimensionality Reduction 14

3.2.6 Feature Importance Assessment 14

3.3 Machine Learning Models 14

3.4 Model Training Process 16

3.5 Validation 17

3.6 Evaluation 19

3.7 Chapter Summary 19

4 Results and Discussion **21**

4.1 Results . . . 21
4.1.1 Experimental Design and Registry . . . 21
4.1.2 Pilot Audit: Reported versus Reproduced Evidence (E0) . . . 23
4.1.3 Phase 2: High Recorded Victimisation (E1, E2) . . . 23
4.1.4 Phase 1: Dominant Crime Type (E3) . . . 25
4.1.5 Class-Imbalance Strategies (E4) . . . 27
4.1.6 Graph Ablations: When Does Message Passing Help? (E5) . . . 28
4.1.7 Feature Ablation and Leakage Check (E6) . . . 29
4.1.8 Robustness over Time, Space, Labels and Resolution (E7–E9) . . . 29
4.1.9 Statistical Uncertainty and Paired Comparisons (E10) . . . 31
4.1.10 Continuous Count Forecasting (E11) . . . 32
4.1.11 Unsupervised Hotspot Detection with DBSCAN (E12) . . . 33
4.1.12 Computational Cost (E13) . . . 34
4.1.13 Post-hoc Extensions: Preprocessing, Significance, Features, Counts and Effective Sample Size (E15–E19) . . . 34
4.1.14 Three-Year Replication (E21) . . . 37
4.1.15 Extension Experiments: Label Definition, Error Disparity and Boosting (E22–E24) . . . 39
4.2 Discussion . . . 41
4.3 Limitations . . . 44
4.4 Application Conversion . . . 45
4.5 Chapter Summary . . . 49

5 Conclusion **50**

References **52**

A Data Dictionary **55**

B Split Manifest and Configurations **57**

C Code Structure and Reproduction **58**

D Learning Outcomes and Requirements **59**

E Traceability to Assessment 1 **60**

F Evidence Inventory and Unresolved Gaps **61**

G Terminal Transcripts of the Runs Performed **62**

List of Figures

3.1 Reproducible pipeline and mapping of stages to research objectives. All models in a phase share labels, features, splits and tuning budget. 9

3.2 Chronological design. Periods 0–2 provide only lagged features; the dates shown are the first days of selected periods. 10

3.3 Occupied cells and the 3 km graph. Panel B shows why inverse-distance weights carry no information on this lattice. 12

3.4 Exploratory analysis: (A) incidents per period; (B) concentration of incidents across cells; (C) Spearman correlation between consecutive periods' cell victimisation counts. 13

3.5 Phase 2 training curves (seed 0). Both models converge; the GCN plateaus at a lower validation AP than the MLP. 17

4.1 Phase 2 test comparison. (A) AP with 95% spatial-block bootstrap intervals; small dots show individual seeds. (B) Macro-F1 under the operational top-20% rule; the dashed line is the RO3 criterion. 24

4.2 Phase 2 test ROC curves (zoomed), precision–recall curves and reliability diagrams. Persistence and KDE are ranking scores, so they have no reliability curve. 25

4.3 Test confusion matrices: Phase 2 under the top-20% rule (LR, ResGCN) and Phase 1 (persistence, LR). 26

4.4 Geographic error analysis on test periods. (A) Phase 2 LR outcomes in period 12. (B) Share of the two test periods in which LR misclassified each cell's dominant type. 26

4.5 Graph ablations. The degradation in performance by the GCN increases as neighbourhood size and layer count increase, while the residual GCN becomes independent of the graph structure, approaching, but not exceeding, the MLP. 28

4.6 Rolling-origin evaluation across five test periods. 30

4.7 Sensitivity of Phase 2 test AP to the label percentile and to grid resolution. . . . 31

4.8 (A) DBSCAN clusters of victimisation incidents in periods 8–10 with flagged cells. (B) PAI on the test periods for DBSCAN zones and for the top-20% cells of each ranking model. 33

4.9 Three-year replication. (A) AP with 95% block-bootstrap intervals. (B) Model-by-model AP in the one-year and three-year studies, close to the diagonal for every model. (C) Rolling-origin AP over the test periods. 38

4.10 Proposed application architecture (design, not deployed). 48

List of Tables

1.1 Final research objectives, changes from Assessment 1 and status 3

2.1 Critical comparison of related work 7

3.1 Objective-to-method mapping 10
3.2 Reconciled cleaning counts (E0) 11
3.3 Phase 2 feature importance on validation periods 9–10 (E14) 14
3.4 Implemented models and validation-selected hyperparameters (P1 = Phase 1, P2 = Phase 2) 15
3.5 Evaluation designs and the generalisation question each answers 19

4.1 Experiment registry (all experiments completed; outputs in `Assessment2/results`) 22
4.2 Assessment 1 pilot evidence (reported) versus reproduction on the same data and split (E0) 23
4.3 Phase 2 test results (periods 11–12; $n = 314$, 66 high). Mean ± sd over 10 seeds where stochastic; 95% CI from the spatial-block bootstrap of seed-averaged scores 24
4.4 Phase 1 test results (periods 11–12; $n = 303$; support: battery 136, criminal damage 27, theft 111, other 29). Mean ± sd over 10 seeds where stochastic . . . 25
4.5 Imbalance strategies for Phase 2 (test). Threshold metrics depend on the decision threshold; AP does not 27
4.6 Graph ablations (test; mean ± sd over 10 seeds). Pilot-style GCN versus residual GCN 28
4.7 Feature-group ablation (test). GCN values are means over five seeds 29
4.8 Rolling-origin evaluation: Phase 2 AP (top) and Phase 1 macro-F1 (bottom) by test period; each model is retrained on all earlier periods 29
4.9 Spatial-block cross-validation for Phase 2: held-out cells' labels masked during training; AP on held-out cells in test periods 30
4.10 Spatial-block bootstrap (26 blocks of 3×3 cells, 2,000 resamples) for paired differences on the test periods 31
4.11 Next-period victimisation count forecasts (test; $n = 314$) 32
4.12 Measured computational cost (Apple M5 CPU; Phase 2; mean over seeds) 34
4.13 Preprocessing ablation (E15, post-hoc). LR is re-tuned per variant on validation; MLP, ResGCN and GCN (5 seeds) keep their E1/E3 hyperparameters 35
4.14 Paired spatial-block permutation tests (E16, post-hoc; 26 blocks, 4,000 permutations; Holm-adjusted within each task) 35

4.15 Phase 1 feature extension (E17, post-hoc): macro-F1 on validation and test periods 36

4.16 Tuned next-period victimisation count forecasts (E18, post-hoc; test periods, n = 314) 36

4.17 Effective sample size and centroid-graph check (E19, post-hoc) 37

4.18 Three-year replication (E21): test periods 33–38 ($n = 948$ cell-periods, 192 high; Phase 1 support 899). Mean ± sd over 10 seeds; CI from the spatial-block bootstrap 38

4.19 Long rolling origin on three-year data (E21): Phase 2 AP and Phase 1 macro-F1 by test period (start date); 5 seeds for the GCN and ResGCN 39

4.20 Phase 2 test results with the narrower FBI index violent-crime label (E22; periods 11–12; 5 seeds for RF and neural models) 40

4.21 Test errors by community-area hardship tercile (E23; 104 test cell-periods per tercile; 95% cell-bootstrap intervals in brackets) 40

4.22 Issues and challenges encountered, responses and status 43

4.23 Running-cost estimate (assumptions: AWS Sydney on-demand Linux prices of 29 October 2026; 730 hours per month; storage, data transfer and tax excluded; prices change and should be re-checked before purchase) 46

4.24 Prototype tests (E20): 20 of 20 passed 48

4.25 Staged implementation plan and feasibility (assumptions: one analyst-developer; existing server; open data) 49

A.1 Model features (computed per cell for target period t from periods $< t$ only) . . . 55

A.2 Raw schema of the one-year CPD extract (17 fields, 229,275 rows) and how each field was used 55

B.1 Period manifest (28-day periods from 24 August 2025) 57

B.2 Sensitivity analysis (E7): Phase 2 test AP / ROC-AUC 57

D.1 Learning-outcome coverage 59

D.2 Assessment requirements mapped to report sections and evidence 59

E.1 Assessment 1 claims and plans checked against the final study 60

F.1 Evidence inventory 61

F.2 Unresolved evidence gaps and the work needed to close them 61

List of Abbreviations

Abbreviation	Meaning	Abbreviation	Meaning
AP	Average precision (step-wise PR-AUC)	MLP	Multilayer perceptron
AUC	Area under the ROC curve	PAI	Predictive Accuracy Index
CI	Confidence interval	PR	Precision–recall
CPD	Chicago Police Department	RBF	Radial basis function
DBSCAN	Density-based spatial clustering	ResGCN	GCN with residual self path
GCN	Graph convolutional network	RF	Random forest
KDE	Kernel density estimation	RO / RQ	Research objective / question
LO	Course learning outcome	SMOTE	Synthetic minority over-sampling
LR	Logistic regression	SVM	Support vector machine
GAT	Graph attention network	MAE, RMSE	Mean absolute / root mean square error

Chapter 1. Introduction

The purpose of this chapter is to restate the research problem as presented in Assessment 1, outline how the problem was refined through the pilot audit process, and define the research objectives, research questions, and report structure

1.1. Problem Statement

Crime is not evenly distributed across a city. A small number of locations account for a substantial proportion of recorded incidents, and this pattern is sufficiently stable to inform planning (Weisburd, 2015). Police and community-safety agencies therefore seek to identify *where* recorded harm is likely to be concentrated during the next planning period and *which type* of crime is likely to predominate in those locations. Traditional hotspot mapping techniques like kernel density estimation (KDE) provide information about locations where past incidents clustered (Chainey, Tompson, & Uhlig, 2008). Machine-learning classifiers can incorporate multiple historical features but treat areas as independent observations. GCNs represent areas as nodes of a spatial graph, allowing each prediction to draw on information from neighbouring areas (Kipf & Welling, 2017; Zubair, Fatima, Ahmed, & Khan, 2025).

In Assessment 1, a two-step approach was proposed using the CPD dataset aggregated in 2.2 km $\times$ 2.2 km cells. Phase 1 involved predicting the predominant crime category in each cell by a GCN and an SVM baseline; Phase 2 involved predicting whether a cell would have high victimisation in the future by a GCN and an LR baseline. The pilot audit of this process prior to conducting the final experiment (Section 4.1.2) uncovered three issues that altered the scope:

1. the Phase 1 features consisted of the same-period crime-type counts whose maximum *is* the label, and hence the task was circular;
2. Phase 2 involved one split between historical and future data with a random 24-cell test set, and hence did not evaluate forecasting into a new period; and
3. a plain LR, as well as a persistence strategy, perfectly replicated the pilot's Phase 2 results on the same split.

The final study thus reformulates both phases as proper one-period-ahead forecasts, employing leakage-free features, matched baselines and chronological testing.

Operational definitions. The prediction unit is a 2.2 km grid cell, and forecasts cover the next 28-day period. The *dominant crime type* is the most frequently recorded primary crime category within a given cell and period; it describes the composition of recorded offending rather than hotspot intensity. *High recorded victimisation* means that a cell ranks among the top 20% of cells for that period based on its count of person-directed incidents: assault, battery, robbery, criminal sexual assault, sex offence, stalking, homicide, kidnapping, offences involving children,

intimidation and human trafficking. These figures count recorded incidents rather than unique victims and are not adjusted for population size. They should therefore not be interpreted as an individual's probability of becoming a victim. The intended users are crime and community-safety analysts developing area-level prevention strategies. The outputs support decision-making and do not provide instructions for automated deployment.

Formally, for cell i in period t, with features $\mathbf{x}_{i,t}$ derived only from earlier periods and graph neighbours $N(i)$, each phase estimates

$$\hat{y}_{i,t} = \arg\max_{y} \; P\Big(Y_{i,t} = y \,\Big|\, \mathbf{x}_{i,t}, \{\mathbf{x}_{j,t}\}_{j \in N(i)}\Big), \qquad \mathbf{x}_{i,t} = f\big(\text{data in periods} < t\big), \tag{1.1}$$

Only graph-based models use the term neighbourhood. The restriction on the use of features from periods before t is the primary methodological change from the pilot.

Existing approaches and products. There are four relevant families of approaches: (i) *Hotspot mapping* (KDE, grids, ellipses) is simple and easy to interpret but only extrapolates historical density (Chainey et al., 2008). (ii) *Conventional machine learning* (LR, SVM, random forest) combines multiple historical features but does not account for neighbouring areas unless this information is explicitly included in the features (Zhang, Liu, Xiao, & Ji, 2020). (iii) *Graph and spatio-temporal networks* learn from neighbouring areas (Han, Hu, Wu, Shen, & Wu, 2020; Kipf & Welling, 2017) but, as this study demonstrates, they do not necessarily outperform strong baselines. (iv) *Deployed predictive-policing products*, such as the grid-based service discussed in Assessment 1, are proprietary; hence, a quantitative comparison with them is not possible. The algorithm of a similar product has been examined for bias (Lum & Isaac, 2016), and its underlying model has been tested in field trials (Mohler et al., 2015). Therefore, this report compares only methods that can be re-run on the same public data.

1.2. Problem Significance

Area-level planning can be justified by the concentration of recorded incidents. In this extract, the top 10% of occupied grid cells contain approximately 30% of incidents, while the top 20% contain about 49% (Figure 3.4B). When available prevention capacity is fixed—for street outreach, lighting and environmental design, victim-support outreach or patrol—accurately ranking areas where recorded harm is expected to concentrate can help use resources more efficiently. As observed in field experiments, forecasts can have an impact on the results since Mohler et al. (2015) discovered that forecasts based on epidemics predicted 1.4–2.2 times more crime than hotspot maps created by analysts and led to a 7.4% decrease in crime volume per unit of patrol time.

The literature also brings out the risks associated with such methods. Recorded crimes represent both the location of the crime and the pattern of police presence. Algorithms based on such crimes will end up emphasizing those neighbourhoods which have heavy police presence (Ensign, Friedler, Neville, Scheidegger, & Venkatasubramanian, 2018; Lum & Isaac, 2016).

Chicago represents a case where predictive algorithms have been developed from data based on policing that is itself contentious (Richardson, Schultz, & Crawford, 2019). The importance of the study is not only about accuracy but also whether the model is better than simpler, auditable models and whether the implementation process has proper checks and balances. The fact that a simple persistence model can capture much of the predictive power is an important one for practical reasons.

The reason for choosing Chicago is that CPD offers geocoded incident level data using a consistent schema. The context surrounding Assessment 1 had New Zealand in mind, but there is no similar data available in coordinates for incident level data; therefore, moving to Auckland or any other city in New Zealand has not yet been attempted.

1.3. Research Objectives

The five objectives identified in Assessment 1 were retained, but they were quantifiable and leak-proof (Table 1.1). The changes that have been made are clearly stated, and Appendix E (Table E.1) shows the process through which each claim and objective in Assessment 1 was developed. Deadlines are not provided because the work was completed within the assessment period.

Table 1.1: Final research objectives, changes from Assessment 1 and status

ID	Final objective (specific and measurable)	Change from Assessment 1	Status
RO1	Build a reproducible pipeline in which the counts from raw to final match exactly, with a documented grid and graph, and features that use only previous periods.	Count reconciliation and history-only rule added.	Achieved (§3.2)
RO2	Predict the next period's dominant crime type; evaluate the performance of GCN versus SVM and matched baselines using macro-F1 metric in held-out periods 11–12.	Same-period features removed; persistence, KDE, LR, RF and MLP added.	Partially achieved (§4.1.4)
RO3	Predict top-20% recorded-victimisation cells; Assessment 1 criterion pre-registered: macro-F1 (mean for both classes) > 0.70 on held-out periods.	Clarified Macro-F1 formula; time-based test replaces random split method.	Achieved; GCN not the best model (§4.1.3)
RO4	Quantify robustness using 10 seeds, spatial-block bootstrap confidence intervals, rolling-origin and spatial-block testing, ablation studies, and sensitivity analysis.	Replaces bootstrap of one fixed pilot test set.	Achieved (§4.1.8)
RO5	Design a usable and governed application and evaluate its practical value and ethical risks.	SHAP is replaced by coefficients and permutation importance.	Design completed and prototype tested, but not deployed (§4.4).

Research questions. The fundamental question is whether a spatial GCN improves area-level predictions of crime type and recorded victimisation more than conventional spatial and machine learning methods.

- **RQ1.** Does a GCN forecast next-period dominant crime type better than a non-graph SVM and other matched baselines?
- **RQ2.** Can history-only features identify top-20% victimisation cells with acceptable minority-class precision and recall, and does the graph add value?
- **RQ3.** How much of the measured skill is real signal rather than chance, given the small number of cells and spatial dependence?

1.4. Structure of the Report

Chapter 2 provides a critique of hotspot, machine-learning, and graph-based methods along with their shortcomings. Chapter 3 describes the implemented method: data, cleansing, grid and graph formation, features, models, learning process, and mathematical details of optimisation for every step. Chapter 4 presents the results, beginning with a pilot audit and twenty-five experiments (E0–E24), results analysis against research objectives, limitations of the current study, and solution design for converting it to an end-user tool. The concluding Chapter 5 includes recommendations for future work. The data dictionary, configurations, splitting manifest, reproducibility steps and learning outcome coverage are given in the appendices. The coverage of the learning outcomes and requirements is tabulated in Appendix D (Tables D.1 and D.2), and the unresolved evidence gaps in Appendix F (Tables F.1 and F.2).

Chapter 2. Literature Review

This chapter updates the Assessment 1 review in the light of what was discovered through experiments, including work on evaluation under dependence, over-smoothing and responsible use of graphs.

2.1. Background

The idea behind place-based crime analysis is that crime occurs in a limited number of places, and this distribution of crime is fairly consistent over time (Weisburd, 2015). Hotspot mapping turns past events into a surface and assumes that the future is similar to the recent past. Chainey et al. (2008) have assessed mapping approaches based on the extent of future events that occur within the mapped hotspots relative to their area, through the Predictive Accuracy Index (PAI), and found KDE to be one of the more consistent methods. Bowers, Johnson, and Pease (2004) have proposed that maps should be prospective, giving greater weight to more recent events because risk diminishes over time and space. This is the standard that any machine learning algorithm would have to surpass: *persistence* (what happened recently continues) and *smoothed density* (what happened nearby continues).

The machine learning approach brings the capacity for combining several signals such as counts, crime mix, arrest and domestic proportions, time of day and learning non-linear interactions (Zhang et al., 2020). The graph neural network framework adds to this by representing which nodes are neighbours. A GCN layer takes the average of each node's features and the features of its neighbours by means of a normalised adjacency matrix and then applies a learned linear transformation (Kipf & Welling, 2017). Li, Han, and Wu (2018) have demonstrated that the procedure is a form of Laplacian smoothing, which is why it works when neighbours have common labels and why stacking layers makes node representations converge (over-smoothing). This theoretical result was crucial to the findings in Chapter 4.

2.2. Related Work

Graph-based crime prediction. Zubair et al. (2025), which is the foundational paper for this research, split the city of Chicago into 2.2 km cells, connected cells within 3 km using inverse-Haversine distances between cell centroids, and trained a two-layer GCN (hidden layers = 128 units, dropout probability = 0.5) to classify crime categories. However, the supplied paper reports accuracies of 78% and F1=0.83, compared to 72% and 0.68 for RBF-SVM and 70% and 0.61 for KDE (the first arXiv version listed 88% accuracy in its abstract; later versions and the paper body report 78%). However, the KDE plot corresponds to the *binary* hotspots task and not to the multiclass task; hence, the two are not directly comparable. The graph is stated to contain 487 nodes and 5,832 edges with an average degree of approximately 11.97, which is possible only when the edges are taken as directed pairs. Whether such a density can be

achieved depends on whether distances are measured between grid centres or incident centroids, which the paper specifies insufficiently: for a regular lattice of grid centres, with a 3 km radius, each node can have at most four neighbours because the distance between diagonal neighbours is 3.11 km (Section 3.2.2), while a centroid-based graph of the 157 cells in this study has an average degree of 4.9 (E19). In addition, the paper does not mention whether its characteristics include crime counts for the same period or how the data were partitioned temporally. These shortcomings do not disqualify the methodology, but they imply that its key figures cannot be used as a benchmark.

Spatio-temporal models. Han et al. (2020) have used an LSTM together with a spatio-temporal GCN for theft count prediction in Chicago communities in sliding windows of time, highlighting the potential of using time-space joint models for counts. Zhang et al. (2020) evaluated the performance of KNN, random forest, SVM, naive Bayes, CNN, and LSTM methods for predicting property crime hotspots in a coastal Chinese city and found that the LSTM method performed best, especially when built-environment covariates (POIs and road density) were included. Both analyses evaluate genuine forecasts of future periods, which the Assessment 1 pilot did not do.

Deployed systems and field evidence. The grid-based forecast products are based on the epidemic-type aftershock model, which was tested in randomised field trials in Los Angeles and Kent by Mohler et al. (2015). Lum and Isaac (2016) applied a similar algorithm to Oakland drug crime data to show that predictions were concentrated in minority neighbourhoods although survey-based estimates of drug use were more evenly distributed; Ensign et al. (2018) have formalised the feedback loops resulting from acting on predictions. Richardson et al. (2019) found that predictive algorithms used by Chicago and other jurisdictions have been trained using datasets generated through illegal or biased practices. ("Strategic Subject List" in Assessment 1 is wrongly associated with Lum and Isaac (2016). This mistake has been corrected here.)

Evaluation under dependence and imbalance. Roberts et al. (2017) have demonstrated that random cross-validation yields an underestimate of error where there are spatial or temporal dependencies in the data and suggested using blocked approaches. For imbalanced binary datasets, precision-recall analysis is better for evaluating performance than ROC analysis (Davis & Goadrich, 2006; Saito & Rehmsmeier, 2015). Class weighting and artificial over-sampling (Chawla, Bowyer, Hall, & Kegelmeyer, 2002) are the usual methods used in the case of imbalanced data, but their usefulness depends on the criterion and the presence of threshold tuning (He & Garcia, 2009).

Table 2.1 presents the most relevant studies and states their implications for this project.

Table 2.1: Critical comparison of related work

Source	Data / unit	Target	Model	Validation	Contribution	Limitation	Implication here
Chainey et al. (2008)	UK incidents; surfaces	Future hotspots	KDE, grid, ellipses	Next-period hit rate, PAI	PAI; KDE consistently useful	No learning	KDE and PAI used as benchmarks
Bowers et al. (2004)	Burglary; points	Near-repeat risk	Prospective weighting	Forward in time	Recency-weighted maps	Single crime type	Recency-weighted persistence baseline
Mohler et al. (2015)	LA, Kent; 150 m boxes	Daily hotspots	ETAS point process	Randomised field trial	Field evidence of benefit	Policing outcomes only	Motivates ranking use case
Zhang et al. (2020)	Chinese city; grid	Property hotspots	KNN, RF, SVM, NB, CNN, LSTM	Held-out data	LSTM best; covariates help	One city, one crime group	Temporal lags; covariates as future work
Han et al. (2020)	Chicago communities	Theft counts	LSTM + ST-GCN	Sliding window	Joint space–time model	Count task, theft only	Count regression added (E11)
Zubair et al. (2025)	Chicago; 2.2 km cells	Crime type	GCN vs SVM, KDE	Split not fully stated	Graph framing of the grid	Graph construction and temporal split not fully specified; KDE on a different task	Base design, re-audited (E0, E5)
Li et al. (2018)	Citation graphs	Node labels	GCN theory	Benchmarks	GCN as Laplacian smoothing	Not spatial	Explains GCN under-performance
Lum and Isaac (2016); Ensign et al. (2018)	Oakland; simulation	Bias, feedback	Simulation, theory	–	Feedback-loop risk	Not a predictive benchmark	Governance in the application

2.3. Research Gaps and Challenges

The revised literature review along with the pilot audit reveals five gaps that are addressed by this study.

1. **Leakage-safe forecasting.** Graph-based crime analysis does not always reveal whether features come before labels. In this study, the history-only principle is imposed and its violation is demonstrated (E0).

2. **Strong, matched baselines.** GCNs have been compared mostly to SVM or KDE; persistence and LR are hardly ever mentioned, even though persistence is the proper baseline for such a concentrated and stable phenomenon. In this case, all methods use the same labels, features, splits and tuning budget.

3. **Evaluation under dependence.** Random divisions of small numbers of cells are not enough to separate out signal from noise. This paper employs chronological analysis, rolling origins, spatial-block cross-validation, and spatial-block bootstrap intervals (Roberts et al., 2017).

4. **Understanding when the graph helps.** Instead of assuming that spatial smoothing

improves performance, the study varies graph radius, weighting and depth, and evaluates a residual GCN inspired by the smoothing analysis of Li et al. (2018).

5. **Recorded victimisation as a distinct task.** Most existing work focuses on either crime types or all-crime hotspots. In this research, the base design is expanded to include recorded offences against persons, clearly stating that these are counts of incidents and not victims.

This paper does not make any claims of novel methodology in graph learning. It makes a careful, reproducible attempt to evaluate whether the published graph design adds value on public data when evaluated fairly.

2.4. Theoretical Framework

Two theoretical constructs underlie the design of the approach. The first one states that *crime concentration and persistence* (Bowers et al., 2004; Weisburd, 2015) imply that the recent counts of cells are good predictors of their future counts, and thus, any learned model should be judged by its improvement over persistence. The second construct, namely *graph smoothing* (Kipf & Welling, 2017; Li et al., 2018), implies that aggregation helps when neighbours have similar labels (homophily) and hinders it when cell-specific signal is strong and labels change sharply between neighbours. Since the 80th-percentile victimisation designation is a highly specific, cell-level ranking construct, the theory would imply that excessive smoothing might decrease accuracy—a prediction which is explicitly tested by the graph ablations in Section 4.1.6.

2.5. Chapter Summary

The literature supports area-level forecasting and warns that published graph results can be hard to reproduce, and that the persistence and density benchmarks are powerful. Dependence and feedback create problems in both evaluation and implementation. The next chapter explains how the research design deals with these issues.

Chapter 3. Methodology

This chapter describes what was actually done. All the code is in the `Assessment2/code` directory (Appendix C); all numbers in Chapters 3–4 are generated by `run_experiments.py` using the provided CSV file.

3.1. Overall Approach

This is a forecasting study. The year of incidents is segmented into thirteen periods of 28 days. Features for each cell are extracted in periods before the target period t, whereas the label describes incidents in t. Models are trained on target periods from 3 to 8, optimised and early stopped on target periods 9–10, and tested once on target periods 11–12, which were not used for any decision (Figures 3.1 and 3.2). The 28-day periods were selected since they comprise four full weeks (thus eliminating any day-of-week composition effect), have 13 periods per year, and match a monthly planning cycle. The last day of the extract (23 August 2026, 22 entries) does not form part of a period and is only considered in the counts. The same procedure was performed on three years of data (24 training, 6 validation and 6 test periods; Section 4.1.14).

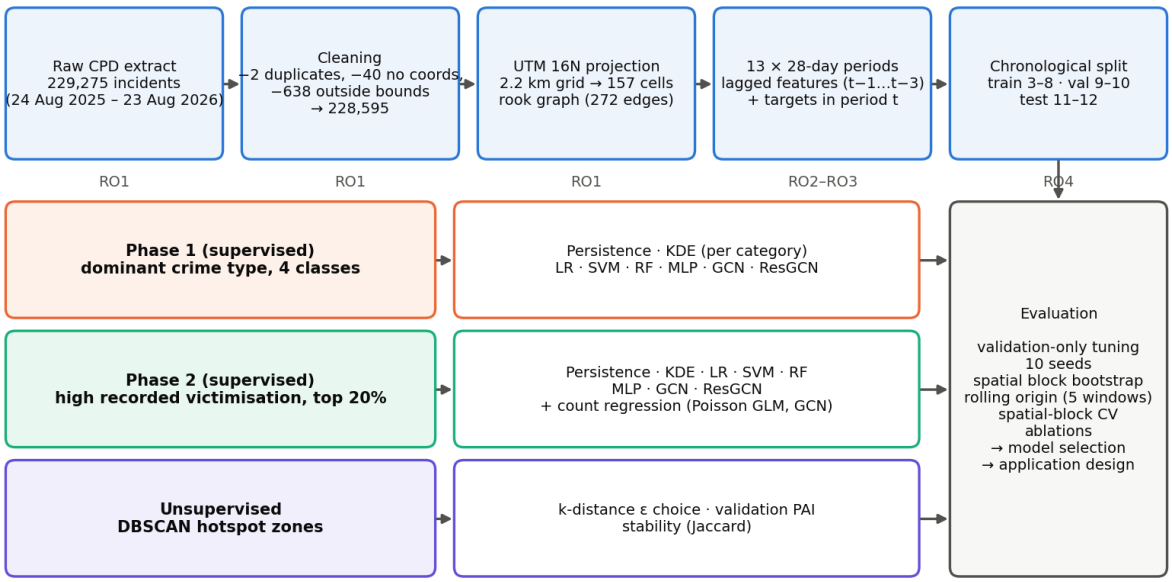


Figure 3.1: Reproducible pipeline and mapping of stages to research objectives. All models in a phase share labels, features, splits and tuning budget.

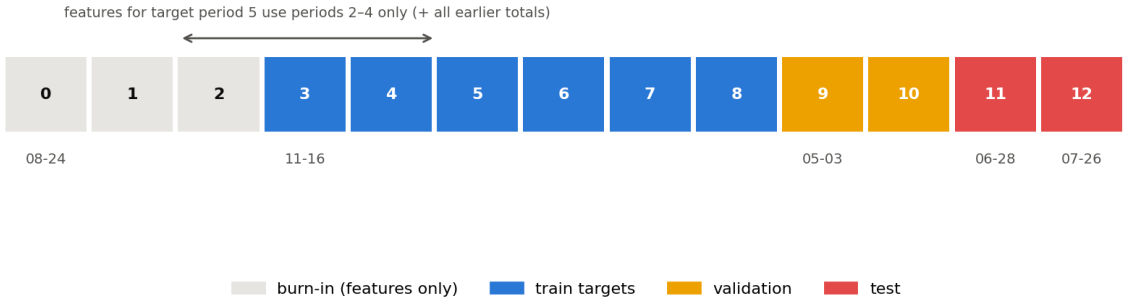


Figure 3.2: Chronological design. Periods 0–2 provide only lagged features; the dates shown are the first days of selected periods.

Table 3.1 links each research objective to its method and supporting evidence.

Table 3.1: Objective-to-method mapping

Objective	Method	Evidence
RO1 pipeline	Stage-by-stage cleaning log; UTM projection; grid; radius graph; history-only feature construction	Table 3.2; Fig. 3.3; E0, E13
RO2 crime type	Persistence, per-category KDE, LR, SVM, RF, MLP, GCN, ResGCN on four classes	E3, E5, E6, E8, E10
RO3 victimisation	Same model set with count regression; top 20% decision rule	E1–E2, E4–E11
RO4 robustness	10 seeds; spatial-block bootstrap; rolling origin; spatial-block CV; sensitivity	E7–E10
RO5 application	DBSCAN hotspot layer; PAI; runtime; architecture and governance design	E12–E14; §4.4

3.2. Data Collection and Preprocessing

Provenance. The data include the CPD *Crimes – One year prior to present* extract from the City of Chicago Data Portal (Chicago Police Department, 2026b), obtained on 31 August 2026 as `Crimes_-_One_year_prior_to_present_20260831.csv` (17 fields; 229,275 rows; 24 August 2025 05:03 to 23 August 2026 00:00). Assessment 1 references the corresponding *Crimes – 2001 to Present* data set, but the filename and field names (such as `CASE#`, `FBI CD`, and `DATE OF OCCURRENCE`) prove that the rolling one-year extract was used, hence the citation has been corrected. The website claims that the data is preliminary, can be reclassified, excludes the last seven days, and is released at block level for confidentiality reasons; all these factors transfer to the limitations (Section 4.3). The data is a non-federal open dataset whose portal licence field reads "See Terms of Use"; the City's terms supply the data "as is" without warranty, allow the City to require use to stop, and require any application built on it to state that the data have been modified from their original source (City of Chicago, 2026c), so the prototype dashboard carries that notice; the files do not include names or narratives but only block-level addresses and coordinates, used here for non-commercial research purposes only. The raw schema and the fields and missingness thereof can be seen in Appendix A (Table A.2). In the three-year replication (Section 4.1.14), the 509,905 records of the same portal's *Crimes – 2001 to Present* database

(Chicago Police Department, 2026a) from 24 August 2023 to 23 August 2025 were collected on 25 October 2026 using the portal's API (17 fields, 106 MB, `Crimes_2001_to_present_subset_20230824_20250823.csv`) and appended to the one-year extract after harmonising field names and arrest and domestic flags.

3.2.1. Data Cleaning

Table 3.2: Reconciled cleaning counts (E0)

Stage	Rule	Remaining	Removed
Raw records	All rows read; all 229,275 timestamps parsed	229,275	–
Exact duplicates	Identical across all 17 fields	229,273	2
Missing coordinates	Latitude/longitude missing (the same 40 rows lack X/Y)	229,233	40
Study boundary	Latitude 41.6–42.1; longitude −87.9 to −87.5	228,595	638

Assessment 1 identified 2 duplicates and 638 boundary exclusions but did not include the 40 records without coordinates, which is why their counts did not add up ($229{,}275 - 2 - 638 = 228{,}635 \neq 228{,}595$). Table 3.2 closes this discrepancy exactly. At this point, all missingness was limited to `LOCATION DESCRIPTION` (1,037 records, 0.45%), set to UNKNOWN and only needed for counting distinct location types, and one missing `WARD`, which is unnecessary. There were 22 duplicate values of `CASE#`, which are kept since one case may lead to multiple offenses. High-frequency cells were not considered outliers, as it was concentration that was modelled. The audit also revealed that the saved Phase 2 file by the pilot contained `HIST_DOMESTIC` and `HIST_ARREST` as concatenated "Y/N" values because the text flags were concatenated as strings instead of summed numerically, while the final pipeline sums the flags after converting them to 0/1 values.

3.2.2. Data Transformation & Feature Engineering

Projection and grid. Coordinates were transformed from WGS84 (EPSG:4326) to UTM zone 16N (EPSG:26916) such that distances are measured in metres. The grid origin is the minimum projected easting and northing of the cleaned data (425,477 m E; 4,610,479 m N), while the cell size is 2,200 m, thus matching the number of cells used by the pilot study, namely 157. Nodes are those cells that are occupied, with each having between 1 and 9,549 incidents (median 1,080). Seventeen cells, which are mostly located near the city borders or lakeshore, have fewer than 50 incidents per year; however, they were retained because their exclusion would bias the ranking towards the centre. Out of 1,570 modelled cell-periods, 68 showed no recorded event in the cell: this is a true zero in the victimisation task but has no dominant type, so it is excluded from the Phase 1 loss and statistics, while remaining a node in the graph.

Graph. Cells whose centres are within 3 km are linked using a KD-tree radius query, resulting in 272 undirected (544 directed) edges, no self-loops stored and no isolated nodes. Since the cells

form a regular lattice, the length of each edge is equal to 2.2 km: the diagonal neighbours have a distance of $2.2\sqrt{2} = 3.11$ km and are out of the radius. The graph is thus a rook-adjacency lattice (degrees 1–4, average 3.46; Figure 3.3), and the weights of the inverse-distance in the proposal are constant $1/2.2 = 0.455$, which GCN normalisation largely cancels. The pilot GCN also did not pass the edge weights to `GCNConv`. Both of these are examined in the graph ablation (E5), which includes graphs at queen (3.2 km), 4.5 km, and 6.6 km. A centroid-based variant, where edge lengths vary (0.68–3.0 km) and inverse distance weights contain information, was also examined (E19, Section 4.1.13).

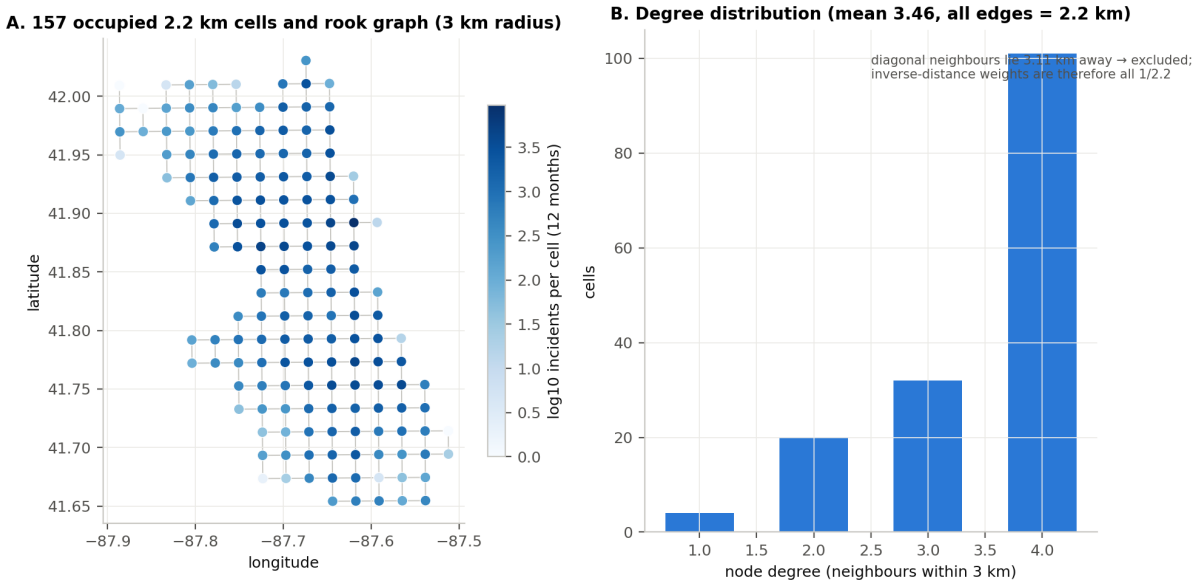


Figure 3.3: Occupied cells and the 3 km graph. Panel B shows why inverse-distance weights carry no information on this lattice.

Targets. In Phase 2, `Y_VICTIM` is the number of person incidents in the cell in period t (73,494 records, 32.2% of the cleaned data; the reasons for category selection are provided in Appendix A). A cell is considered *high* if it has an incident count equal to or greater than the 80th percentile in that period among the 157 cells. Ties at the cutoff result in prevalence between 20.4 and 22.3% per period (32–35 cells). Percentiles are used only retrospectively to define labels; when predicting, the program ranks the cells and marks the best 32 (20%). No future outcome is needed to make the prediction, and no held-out outcome is used to choose a threshold.

In Phase 1, categories with fewer than 1,000 instances were collapsed to OTHER (Pilot Rule), the dominant category was calculated for each cell period (ties resolved in favour of the more frequent one), and those categories that were dominant in at least 5% of *training* cell periods were selected as classes. The training data only retained THEFT (42.5%), BATTERY (38.8%), and CRIMINAL DAMAGE (7.0%) classes; all other categories were dominant in no more than 2.6% of cell periods and were collapsed to OTHER_TYPE. The collapsing eliminates the singleton class (one SEX OFFENSE cell) that prevented stratified folds in the Pilot case.

Features. A total of twenty-three features were created for each cell/target combination for earlier periods only (Appendix A, Table A.1). These include victimisation counts and total counts at lags 1 to 3, long-run means of all previous periods, the victimisation, domestic, arrest and night-time shares of all previous incidents, crime mix of three periods (share of eight common types), trend for one to three periods (lag 1 minus lag 3), and coordinates of the cell centre. Counts were log-transformed with $\log(1 + x)$, and the trend with the signed logarithm since counts have strong right-skewness (cell means span from 0 to over 150 per time period). All features were then standardised using the mean and standard deviation from only the training rows. Two neighbour average features were constructed using the row-normalised adjacency matrix only for the spatial lag ablation study.

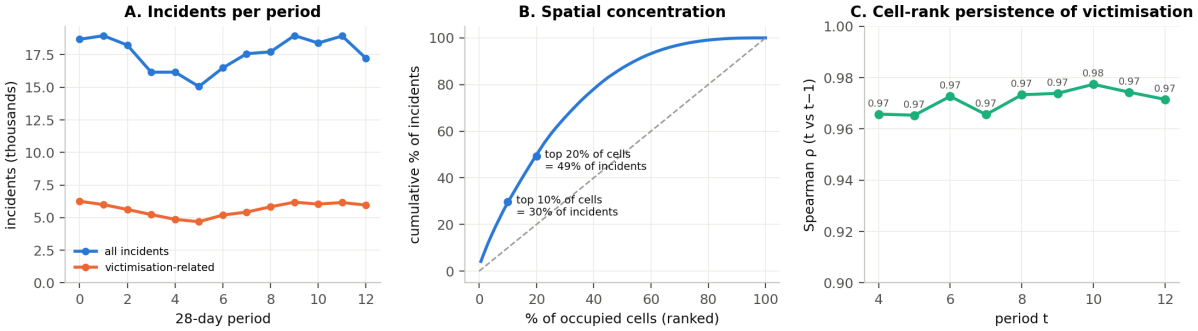


Figure 3.4: Exploratory analysis: (A) incidents per period; (B) concentration of incidents across cells; (C) Spearman correlation between consecutive periods' cell victimisation counts.

Figure 3.4 highlights some insights gained from exploratory analysis that guided the design process. There is a dip in volumes during the winter periods (4-5) and an increase during the summer periods, hence the choice of a relative per period label based on the ranking as opposed to an absolute count threshold. Concentration (Panel B) supports aggregation at the area level. Most importantly, Panel C reveals that ranking cells in terms of victimisation does not change much between periods (Spearman $\rho = 0.965$-0.977), so rankings are more stable than raw counts, which predicts that persistence will be a very strong baseline.

3.2.3. Class Imbalance Resolution

The Phase 2 dataset has a 1:4 imbalance by construction. Three approaches were compared using identical splits (E4): no correction, inverse-frequency class weights, and SMOTE (Chawla et al., 2002; Lemaître, Nogueira, & Aridas, 2017) applied only to the training rows of the tabular LR model. Each approach was evaluated at a 0.5 threshold and at a threshold selected using validation data. SMOTE was not applied to graph-based models because synthetic nodes would require invented edges and coordinates without spatial meaning. Instead, class-weighted cross-entropy (Equation 3.3) was used for the GCN, ResGCN and MLP. The main decision rule—flagging the 32 highest-ranked cells in each period—matches the prevalence by construction and is unaffected by probability calibration. For Phase 1 (test class shares of 45/9/37/10%), all models used balanced class weights, and macro-F1 served as the selection metric, giving every class equal importance, including the less common CRIMINAL DAMAGE and OTHER_TYPE classes

3.2.4. Feature Selection

No automatic feature selection was employed. Given 23 features and 942 training cell-periods, L2 regularisation in LR and weight decay in neural models stabilize variance better than feature selection, which itself requires validation data. Feature group ablations (E6) evaluate if victimisation history, crime mix proportions, and coordinates are required, as well as whether removing composition features changes Phase 1, given that same-period composition features had caused leakage in the pilot.

3.2.5. Dimensionality Reduction

Dimensionality reduction was not considered necessary. The space of the features is low-dimensional, the features are meaningful and partially correlated by design (victimisation counts form a part of the total counts), and the use of principal components would destroy the interpretability that is required for the explanation module of the application. Collinearity is addressed using regularisation and is acknowledged when interpreting coefficients.

3.2.6. Feature Importance Assessment

Importance was computed only on the validation data (E14), replacing the planned SHAP analysis in Assessment 1, as, for a linear model, coefficients and permutation importance provide an adequate explanation. The six largest LR coefficients are listed in Table 3.3.

Table 3.3: Phase 2 feature importance on validation periods 9–10 (E14)

Feature	LR standardised coefficient	LR permutation ΔAP	RF permutation ΔAP (± sd)
Mean victimisation, all previous periods	+5.64	0.267	0.057 ± 0.027
Total incidents, lag 1	+4.16	0.142	0.007 ± 0.011
Mean total incidents, all previous periods	+3.99	0.129	0.007 ± 0.011
Victimisation share of all previous incidents	+3.99	0.214	0.007 ± 0.005
Assault share, last 3 periods	−2.68	0.035	-0.001 ± 0.003
Battery share, last 3 periods	−2.31	0.030	0.006 ± 0.005

Long-run victimisation mean is the top variable in both models: shuffling it decreases validation AP by 0.27 for LR. Then come recent total volume and historical victimisation share. The negative coefficients on the assault and battery shares depend on the victimisation mean already in the model and thus should not be interpreted causally. RF distributes feature importance among correlated counts, so its individual decrease is modest. This ranking guided the choice of features in the explanation panel for the application (Section 4.4) and the "victimisation lags only" ablation (E6).

3.3. Machine Learning Models

Models were selected to distinguish four types of skill: persistence, spatial averaging, flexible table-based learning, and graph learning. The GCN and non-graph MLP have the same depth, width options and training code, so that any difference in their behaviour reflects the effect

of message passing. All used models are listed in Table 3.4 together with hyperparameters optimised on the validation dataset.

Table 3.4: Implemented models and validation-selected hyperparameters (P1 = Phase 1, P2 = Phase 2)

Model	Specification	Selected hyperparameters	Role
Majority	Predict majority class; AP equals prevalence	–	Null reference
Persistence	P2 score $= 1 \cdot L_1 + 0.5 \cdot L_2 + 0.25 \cdot L_3$ (recency-weighted victimisation lags); P1 dominant category of last three periods	–	Concentration theory
KDE	Gaussian KDE of incidents in previous three periods, evaluated at cell centres; P1 uses one count-scaled KDE per category, argmax	Bandwidth 800 m (P1 and P2)	Traditional hotspot method
LR	L2-regularised logistic / multinomial regression, balanced class weights	$C = 10$ (P2); $C = 100$ (P1)	Interpretable baseline
SVM (RBF)	Kernel SVM, balanced weights; decision function used for ranking	$C = 10, \gamma = 0.01$ (P2); $C = 0.1$, γ=scale (P1)	Proposal baseline
Random forest	300 trees, balanced-subsample weights (Breiman, 2001)	Unlimited depth, leaf 1 (P2); leaf 5 (P1)	Non-linear tabular baseline
MLP	Two linear layers, ReLU, dropout; same code path as GCN without edges	$h = 128, p = 0.3, \eta = 0.01, \lambda = 5 \times 10^{-3}$ (P2); $h = 32, p = 0.3, \eta = 0.003$ (P1)	Graph ablation
GCN	Two `GCNConv` layers with self-loops, symmetric normalisation and edge weights (Kipf & Welling, 2017)	$h = 64, p = 0.3, \eta = 0.01, \lambda = 5 \times 10^{-4}$ (P2); $h = 64, p = 0.5, \lambda = 5 \times 10^{-3}$ (P1)	Proposal primary model
GAT	Two `GATConv` layers, 4 attention heads, ELU; learned neighbour weights, edge weights unused (Veličković et al., 2018)	$h = 64, p = 0.3, \eta = 0.01, \lambda = 5 \times 10^{-4}$ (P2); $h = 128$ (P1)	Learned-neighbourhood test
ResGCN	GCN layer plus separate linear self path in each layer (Equation 3.2)	$h = 128, p = 0.3, \eta = 0.003, \lambda = 5 \times 10^{-3}$ (P2); $h = 64, p = 0.3$ (P1)	Optimisation from smoothing analysis
DBSCAN	Density clustering of victimisation incident coordinates (Ester et al., 1996)	$\varepsilon = 300$ m, minPts $= 60$	Unsupervised hotspot layer
Poisson GLM; GCN-Poisson	Next-period count regression with log link	GLM $\alpha = 1$	Continuous score (E11)

GCN propagation. With weighted adjacency A, self-loops $\tilde{A} = A + I$ and degree matrix $\tilde{D}$, one layer computes

$$H^{(l+1)} = \sigma\left(\tilde{D}^{-1/2}\tilde{A}\tilde{D}^{-1/2}H^{(l)}W^{(l)}\right). \tag{3.1}$$

For node i with degree d_i on an equally weighted graph, the weight on its own features is $1/(d_i + 1)$ and on neighbour j it is $1/\sqrt{(d_i + 1)(d_j + 1)}$. For this rook lattice (average degree 3.46), the weight on its own history for a typical cell is therefore only about 0.2 after one layer, and even less after two. With the inverse-distance weights being constant, scaling of the matrix A by 1/2.2 affects only the self-weight through the self loop, which explains why the weighted and unweighted graphs behave almost identically (E5). For a target where the rank is that of the *own* cell, this process blends a highly victimised cell with its less victimised neighbouring cells. This analysis led to two decisions, made before test evaluation and based on validation results: to investigate the depth and graph radius, and to include a residual self path,

$$H^{(l+1)} = \sigma\left(\hat{A}H^{(l)}W_n^{(l)} + H^{(l)}W_s^{(l)}\right), \qquad \hat{A} = \tilde{D}^{-1/2}\tilde{A}\tilde{D}^{-1/2}, \tag{3.2}$$

so that the model is able to learn how much neighbour data it needs instead of having this determined through node degree. The model resembles the self/neighbour separation of GraphSAGE-

style architectures and was implemented using PyTorch and PyTorch Geometric (Fey & Lenssen, 2019; Paszke et al., 2019).

Loss and regularisation. Classification networks minimise class-weighted cross-entropy over labelled nodes V_L,

$$\mathcal{L} = -\frac{1}{|V_L|} \sum_{i \in V_L} w_{y_i} \log \hat{p}_{i,y_i}, \qquad w_c = \frac{n}{K\, n_c}, \tag{3.3}$$

where n_c represents the number of training nodes belonging to class c out of K classes. The weight increases the gradient contribution for minority classes, which is cost-sensitive learning due to imbalance (He & Garcia, 2009). Adam with weight decay λ applies L2 regularisation by adding $\lambda\|W\|^2$, which is important because there are 157 nodes per snapshot.

Logistic regression and SVM. LR models the log-odds as linear in the standardised features and minimises

$$\min_{\beta} \ \tfrac{1}{2}\|\beta\|^2 + C \sum_i w_{y_i} \log\!\left(1 + e^{-\tilde{y}_i \beta^\top \mathbf{x}_i}\right), \qquad P(y = 1 \mid \mathbf{x}) = \frac{1}{1 + e^{-\beta^\top \mathbf{x}}}, \tag{3.4}$$

with $\tilde{y}_i \in \{-1, 1\}$, so higher C implies weaker regularisation. The validation search selected $C = 10$ for the model using 23 standardised predictors. Validation AP ranged from 0.964 to 0.968 across the tested values of C, indicating that performance was relatively insensitive to this choice and that the 23 standardised features are not over-fitted at this sample size. The RBF-SVM maximises the margin in the feature space defined by $k(\mathbf{x}, \mathbf{x}') = \exp(-\gamma\|\mathbf{x} - \mathbf{x}'\|^2)$ (Cortes & Vapnik, 1995); the small $\gamma = 0.01$ selected for Phase 2 produces a smooth decision boundary, described here as nearly linear and consistent with the LR result.

KDE. Density in the cell centre s from n incidents s_j occurring in the past three periods is

$$\hat{f}(s) = \frac{1}{2\pi n h^2} \sum_{j=1}^{n} \exp\!\left(-\frac{\|s - s_j\|^2}{2h^2}\right). \tag{3.5}$$

Bandwidth h controls the trade-off between bias and variance (Silverman, 1986). Validation AP peaked at 0.936 when $h = 800$ m but decreased to 0.774 at $h = 3,000$ m. The wider kernels blurred neighbouring high- and low-victimisation cells, producing the same over-smoothing problem observed in the GCN.

3.4. Model Training Process

Splits. Unit of observation is a cell-period. Training data uses $6 \times 157 = 942$ rows for Phase 2 (919 non-zero rows for Phase 1), validation uses 314, and the test set uses 314 (Phase 2) and 303 (Phase 1). The observations are not independent: the same 157 cells appear repeatedly and neighbours are correlated, thus the sample size is much less than the number of incidents (228,595) and is closer to the number of cells. Chronological splitting has been selected over

random node splitting in order to ensure that each test prediction is an actual forecast; the two-period test window gives 314 test rows compared to 24 cells in the pilot.

Graph protocol. Each period constitutes a separate snapshot of the same graph, and the snapshots are combined into a block-diagonal graph so that messages do not pass between periods. The protocol is *inductive in time*: all node features, derived only from historical information, and all edges are available at inference, while no labels from the target period are used. In the spatial-block experiment (E9), the protocol is *transductive in space*: held-out cells remain in the graph with their historical features, but their labels are excluded from the loss calculation and early stopping.

Optimisation and convergence. Neural models were trained using full-batch Adam for up to 400 epochs, with early stopping (patience 60) based on validation AP (Phase 2) or macro-F1 (Phase 1), restoring the best checkpoint. Tabular models were fitted using scikit-learn (Pedregosa et al., 2011). Training is fast: a GCN converges in approximately 0.45 s and LR in approximately 4 ms on the CPU (Table 4.12). Figure 3.5 shows that the GCN's validation AP increases rapidly before levelling off below the MLP's, suggesting architectural bias rather than insufficient training.

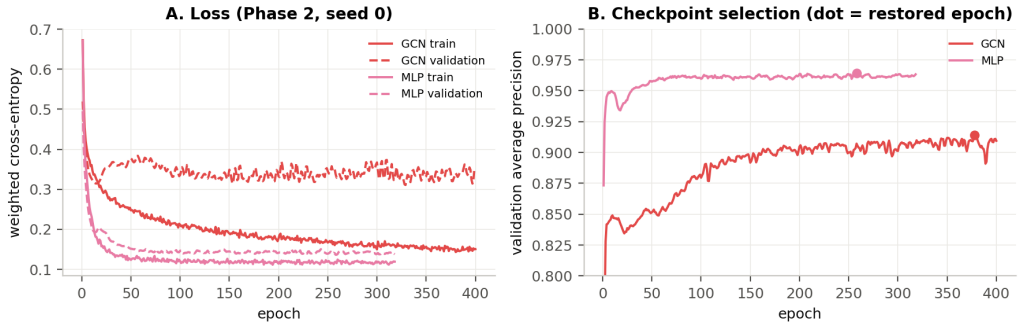


Figure 3.5: Phase 2 training curves (seed 0). Both models converge; the GCN plateaus at a lower validation AP than the MLP.

Reproducibility. Software: Python 3.9.6, NumPy 2.0.2, pandas 2.3.3, scikit-learn 1.6.1, PyTorch 2.8.0, PyTorch Geometric 2.6.1, pyproj 3.6.1 and imbalanced-learn 0.12.4 on macOS (Apple M5, 10-core, 16 GB RAM). All stochastic models used seeds 0–9. As the split is determined exclusively by the dates (Appendix B, Table B.1), no random-split manifest is needed. The entire suite of experiments takes less than ten minutes of compute time with maximum memory usage of 0.94 GB.

3.5. Validation

Hyperparameter optimisation. In Assessment 1, the base paper's hyperparameters were followed, but with a promise of further optimisation to come. This was reconciled with a validation-only approach: each model had a grid search performed over periods 9–10, neural

networks were optimised over three seeds, and the test periods were only used once, after all decisions were made. The grid searches consisted of the following: LR $C \in \{0.01, 0.1, 1, 10, 100\}$; SVM $C \in \{0.1, 1, 10, 100\} \times \gamma \in \{\text{scale}, 0.01, 0.1\}$; RF depth $\in \{3, 6, \text{none}\}\times$ minimum leaf $\in \{1, 5\}$; MLP/GCN/ResGCN hidden $\in \{32, 64, 128\}\times$ dropout $\in \{0.3, 0.5\}\times$ learning rate $\in \{0.01, 0.003\}\times$ weight decay $\in \{5 \times 10^{-4}, 5 \times 10^{-3}\}$ (24 configurations $\times$ 3 seeds); KDE bandwidth $\in \{300, 500, 800, 1200, 2000, 3000\}$ m. The depth and graph radius were then tested as ablations (E5) and were *not* used for selecting the reported models. Model selection was performed according to the stated rule: the best validation performance in each phase, where the simpler model was selected only if the performances differed by less than one seed standard deviation. The Assessment 1 suggested a smaller grid (learning rate $\{0.001, 0.01\}$, hidden width $\{64, 128\}$, dropout $\{0.3, 0.5\}$); the final grid includes these widths and dropouts, uses learning rates $\{0.01, 0.003\}$ (0.001 was not considered), and incorporates weight decay and widths of 32 for the neural models, hence it is a superset, except for the learning rate.

Over- and under-fitting. The highest validation AP achieved by the GCN (0.910) was lower than that of any tabular method and the MLP (0.962–0.968), with low seed variation (sd ≤ 0.004). Therefore, the gap is architectural, not caused by overfitting; having more neurons or reducing the dropout ratio did not narrow the gap. The pilot GCN configuration (128 units) was not chosen for either phase. Weight decay with a value of 5×10^{-3} was favoured by both the MLP and ResGCN models consistent with the small training graph.

Robustness designs. Four additional designs address different generalisation questions: (i) ten training seeds assess randomness in the optimisation process; (ii) a *rolling origin* (E8) retrains on all preceding periods before each test period 8–12, assessing stability over time; (iii) *spatial-block cross-validation* (E9) holds out five k-means blocks of cells to assess transfer to unseen areas; and (iv) a *spatial-block bootstrap* (E10) resamples 26 blocks of 3×3 cells (6.6 km squares) with replacement 2,000 times, keeping both test periods of each block together, to obtain confidence intervals and paired model differences that account for spatial dependence (Roberts et al., 2017). Stratified five-fold cross-validation of cells, proposed in Assessment 1, was not used because it would mix periods (temporal leakage) and because the rare Phase 1 classes cannot appear in every fold. Nested cross-validation was considered unnecessary because the validation-then-test procedure already separates model tuning from evaluation. However, the selected configurations may have optimistic validation scores, so conclusions are based only on test scores.

Table 3.5 summarises the evaluation designs and the generalisation question each answers.

Table 3.5: Evaluation designs and the generalisation question each answers

Design	Question answered	Used here
Random node split	Can the model interpolate between cells it has seen? (optimistic when cells and periods are correlated)	Assessment 1 pilot only; replaced
Chronological hold-out	Can it forecast a future period from the past?	E1–E3 (periods 11–12); E21 (33–38)
Rolling origin	Is forecasting skill stable as the origin moves, and across seasons?	E8 (5 windows); E21 (12 windows)
Spatial-block hold-out	Can it transfer to areas whose labels it never saw?	E9 (5 k-means blocks, labels masked)
Spatial-block bootstrap / permutation	How uncertain are differences given spatial dependence?	E10, E16 (3×3-cell blocks)

3.6. Evaluation

Phase 2 is a ranking task with 21% prevalence. The main metric is **average precision (AP)**, the step-wise area under the precision–recall curve calculated using scikit-learn (not trapezoidal PR-AUC), which reflects minority-class performance (Saito & Rehmsmeier, 2015). The decisions are measured with the practical **top-20% rule** (select 32 cells per period), measuring positive-class F1 and macro-F1 (the average of the high-class and low-class F1; this is the metric used in the RO3 criterion). Metrics at a validation-tuned probability threshold, ROC-AUC and the Brier score are secondary. For Phase 1, the main metric is macro-F1 for the four classes, with accuracy, weighted F1, balanced accuracy, Cohen's κ and per-class F1. Following Sokolova and Lapalme (2009),

$$\text{Precision} = \frac{TP}{TP+FP}, \quad \text{Recall} = \frac{TP}{TP+FN}, \quad F_1 = \frac{2PR}{P+R}, \quad \text{Accuracy} = \frac{TP+TN}{N}, \tag{3.6}$$

$$\text{macro-}F_1 = \frac{1}{K}\sum_{c=1}^{K} F_{1,c}, \qquad \text{weighted-}F_1 = \sum_{c=1}^{K} \frac{n_c}{N} F_{1,c}, \qquad \text{AP} = \sum_{k} (R_k - R_{k-1}) P_k, \tag{3.7}$$

$$\text{Brier} = \frac{1}{N}\sum_{i} (\hat{p}_i - y_i)^2, \qquad \text{PAI} = \frac{n_{\text{hot}}/n}{a_{\text{hot}}/a}. \tag{3.8}$$

ROC-AUC is the probability that a randomly selected high cell is ranked above a randomly selected low cell. The null references are different for each measure: the AP for a random ranking is the prevalence (0.21), the macro-F1 of the all-low classifier is 0.44, and Phase 1 majority accuracy is 0.37 with macro-F1 0.13. Thus, 0.5 is not a universal chance line. For the hotspot perspective, the PAI (Chainey et al., 2008) is the ratio of the percentage of victimisation events from the test period to the percentage of cells selected. For count regression (E11), a truly continuous problem, MAE, RMSE, R^2 and Spearman ρ are used.

3.7. Chapter Summary

The methodology transforms the pilot's random and partially circular design into a chronological and history-only forecasting design; aligns labels, features, and tuning budgets for nine model families; settles the hyperparameter dilemma with a validation-only search; and gauges uncertainty with blocked designs. The mathematics of graph normalisation and kernel band-

width influenced two specific design decisions—the residual GCN and the graph and bandwidth searches—whose impact is evaluated in the next chapter.

Chapter 4. Results and Discussion

This chapter first distinguishes the Assessment 1 pilot evidence from the new final experiments before presenting their results. These results are interpreted in relation to the research questions and objectives, followed by an analysis of limitations and challenges and a design for converting the work into an application. Unless specified otherwise, "test" refers to periods 11–12 (28 June to 22 August 2026), which were used only once after all choices had been finalised.

4.1. Results

4.1.1. Experimental Design and Registry

All experiments are presented in Table 4.1. Experiments E0–E14 and E20–E21 adhere to the design determined before analysis of the test periods; experiments E15–E19 and E22–E24 are defined as *post-hoc*, since they were designed after the main test results were obtained, to fill gaps revealed during the review of the report; thus any choice in them is based on validation data exclusively and cannot influence the chosen models from Sections 4.1.3–4.1.4. All experiments finished. Tuning budget (validation only): LR 5 values of C; SVM 12 settings; RF 6; KDE 6 bandwidths; and for each neural model (MLP, GCN, ResGCN, GAT) 24 configurations $\times$ 3 seeds; the final models used 10 seeds, while the ablation and sensitivity studies used 5. Each varies only one parameter from the primary baseline and uses the same splits, labels and tuning budget. "Seeds" refers to how many training seeds were tried on stochastic models.

Table 4.1: Experiment registry (all experiments completed; outputs in `Assessment2/results`)

ID	Question	Design / changed factor	Models	Metric	Seeds
E0	Are pilot results valid?	Reconcile counts; leakage rule; reproduce pilot split with LR	Pilot GCN (reported), LR, persistence	Accuracy, macro-F1, AP	–
E1	Best hyperparameters?	Validation-only grid search (periods 9–10)	All	Val. AP / macro-F1	3
E2	P2 test comparison	Chronological test, periods 11–12	8 models + majority	AP, macro-F1, AUC, Brier	10
E3	P1 test comparison	Same design, four classes	8 models + majority	Macro-F1, κ, per-class F1	10
E4	Does imbalance handling help?	None / class weights / SMOTE; 0.5 vs tuned threshold	LR, GCN	Threshold F1, AP	10
E5	Does the graph help?	Radius, weighting, depth; spatial-lag features	GCN, ResGCN, LR	AP, macro-F1	10
E6	Which features matter? Leakage?	Remove feature groups	LR, GCN	AP, macro-F1	5
E7	Sensitive to label or grid?	70/80/90th percentile; 1.5/2.2/3.0 km cells	Persistence, LR, GCN, ResGCN	AP, AUC	5
E8	Stable over time?	Rolling origin, test periods 8–12	Persistence, KDE, LR, GCN, ResGCN	AP, macro-F1	5
E9	Transfer to unseen areas?	Five spatial blocks held out (labels masked)	Persistence, LR, MLP, GCN, ResGCN	AP, AUC	5
E10	Signal or chance?	Spatial-block bootstrap (26 blocks, 2,000 reps); paired differences	All	AP, macro-F1 CIs	–
E11	Can counts be forecast?	Continuous next-period count	Persistence, Poisson GLM, GCN, MLP	MAE, RMSE, R^2	5
E12	Unsupervised hotspots useful?	DBSCAN grid; validation PAI; stability	DBSCAN vs ranked models	PAI, hit rate	–
E13	Cost?	Fit/inference time, memory, parameters	All	Seconds, MB	–
E14	Why does it predict?	Coefficients, permutation importance (validation)	LR, RF	ΔAP	–
E15	Does preprocessing help? (post-hoc)	Raw / z-score / log / log+z-score / robust scaling; drop low-volume cells	LR, MLP, GCN, ResGCN	AP, macro-F1	5
E16	Are differences real? (post-hoc)	Paired spatial-block permutation tests, Holm-corrected	All	AP, macro-F1	–
E17	Can Phase 1 features improve? (post-hoc)	+18 history-only features, validation-selected	LR, RF, MLP, ResGCN	Macro-F1	5
E18	Are count models fairly tuned? (post-hoc)	Tuned neural count models; α grid to 100	GLM, MLP, GCN, ResGCN, GAT	MAE, RMSE, R^2	5
E19	Effective sample size; weighted graph? (post-hoc)	Moran's I, lag-1 correlation; centroid-based graph	GCN, ResGCN, GAT	n_{eff}, AP	10
E20	Does the prototype work?	20 automated API tests, serving parity, latency	LR champion	Pass/fail, ms	–
E21	Do findings hold over three years?	24 train / 6 validation / 6 test periods; 12-window rolling origin	All	AP, macro-F1	10
E22	Does a narrower victimisation label change the ranking? (post-hoc)	FBI index violent crime only; re-tuned on validation	Persistence, KDE, LR, RF, MLP, GCN, ResGCN	AP, macro-F1	5
E23	Are errors spread evenly across poorer and richer areas? (post-hoc)	Test errors by community-area hardship tercile; cell bootstrap	LR, persistence, RF, MLP, GCN, ResGCN	Recall, FPR, flag rate, macro-F1	–
E24	Does gradient boosting add skill? (post-hoc)	Histogram gradient boosting, 8-setting validation grid	HGB	AP, macro-F1	5

4.1.2. Pilot Audit: Reported versus Reproduced Evidence (E0)

Table 4.2 contrasts what was *reported* in the proposal with what this study *replicated*. The label of the Phase 1 pilot is the arg-max of 17 columns of crime-type counts, which were also the input features; applying the arg-max rule to the pilot feature file yields the classification label of all 157 grid cells (1.000 accuracy). Hence, the pilot task was self-referential, and the 66.67% GCN accuracy on the pilot measured performance at learning an arg-max from the raw count data and not forecasting skill. In Phase 2, replicating the pilot's stratified random split (seed 42; 109/24/24 grid cells; 5 positive test cells) and training the class-weighted LR on the same ten features yielded perfect scores; ranking cells by historical victimisation rate, with the threshold set at the training-set 80th percentile, was also perfect. The Spearman correlation between past and future victimisation rates across the entire 157 cells was 0.989. Therefore, the pilot study's 0.842 macro-F1 was inferior to simpler and clearer alternatives, and with only five positive test cells, each false negative changed the recall score by 20 percentage points. The statement in the proposal that bootstrap resampling disproved luck in the split is withdrawn: resampling the same 24 test cells cannot reflect any variance in splitting or training.

Table 4.2: Assessment 1 pilot evidence (reported) versus reproduction on the same data and split (E0)

Item	Reported in Assessment 1	Reproduced / audited in this study
Cleaning counts	229,275 → 228,595 (2 duplicates, 638 boundary)	Reconciled: also 40 missing coordinates (Table 3.2)
Phase 1 GCN (7 classes)	Acc. 66.67%, macro-F1 0.218	Label = arg-max of input features: rule accuracy 1.000 on 157 cells (circular task)
Phase 1 SVM	Acc. 37.50%, macro-F1 0.134	Same circularity; not comparable with forecasting
Phase 2 GCN (24 test cells, 5 positive)	Acc. 0.875, macro-F1 0.842, AUC 0.989, PR-AUC 0.967	Not re-run; saved file had string-valued arrest/domestic columns
Phase 2 LR, same split	Not run	Acc. 1.000, macro-F1 1.000, AUC 1.000, AP 1.000
Phase 2 persistence	Not run	Acc. 1.000 with training-set threshold
Graph (3 km, inverse distance)	157 nodes, 544 directed edges	Confirmed; all edges 2.2 km, so weights are constant

4.1.3. Phase 2: High Recorded Victimisation (E1, E2)

Table 4.3 and Figure 4.1 report the main Phase 2 comparison on 314 test cell-periods (66 high). Every non-trivial model exceeded the pre-registered RO3 criterion (macro-F1 > 0.70) by a wide margin, but so did persistence (0.932) and KDE (0.932), which shows that the criterion set in Assessment 1 was too weak to discriminate between models. The informative comparisons are therefore against persistence. LR, RF and the MLP formed the top group (AP 0.980–0.988; top-20% macro-F1 0.951–0.952). The pilot-style GCN was clearly worst among the learned models (AP 0.885 ± 0.023, macro-F1 0.860 ± 0.013) and the most variable across seeds. The residual GCN recovered almost all of the gap (AP 0.975 ± 0.001) but did not exceed persistence by more

than noise. The graph-attention network (GAT), which learns its own neighbour weights, did no better than the GCN (AP 0.867 ± 0.020): attention lets a node reweight its neighbours, but without a separate self path it cannot restore the diluted own-cell signal. Section 4.1.13 adds paired permutation tests, and Section 4.1.14 repeats the whole comparison on three years of data.

Table 4.3: Phase 2 test results (periods 11–12; $n = 314$, 66 high). Mean ± sd over 10 seeds where stochastic; 95% CI from the spatial-block bootstrap of seed-averaged scores

Model	AP	AP 95% CI	ROC-AUC	Top-20% rule		Tuned-threshold	Brier
				Pos. F1	Macro-F1	macro-F1	
Majority (all low)	0.210	–	0.500	0.000	–	0.441	–
Persistence	0.973	[0.938, 0.994]	0.993	0.892	0.932	–	–
KDE ($h = 800$ m)	0.970	[0.934, 0.993]	0.992	0.892	0.932	–	–
LR	**0.983**	[0.954, 0.997]	0.996	0.923	0.951	0.907	0.028
SVM (RBF)	0.974	[0.939, 0.993]	0.993	0.892	0.932	0.925	–
RF	0.988 ± 0.001	[0.968, 0.999]	0.997	0.923	0.951	0.944 ± 0.010	0.023
MLP	0.980 ± 0.001	[0.954, 0.996]	0.994	0.925 ± 0.005	0.952 ± 0.003	0.938 ± 0.007	0.040
GCN	0.885 ± 0.023	[0.814, 0.944]	0.957 ± 0.012	0.778 ± 0.021	0.860 ± 0.013	0.866 ± 0.013	0.080
ResGCN	0.975 ± 0.001	[0.954, 0.992]	0.993	0.888 ± 0.007	0.929 ± 0.005	0.935 ± 0.005	0.037
GAT	0.867 ± 0.020	[0.803, 0.950]	0.958 ± 0.006	0.785 ± 0.013	0.864 ± 0.008	0.867 ± 0.017	0.109

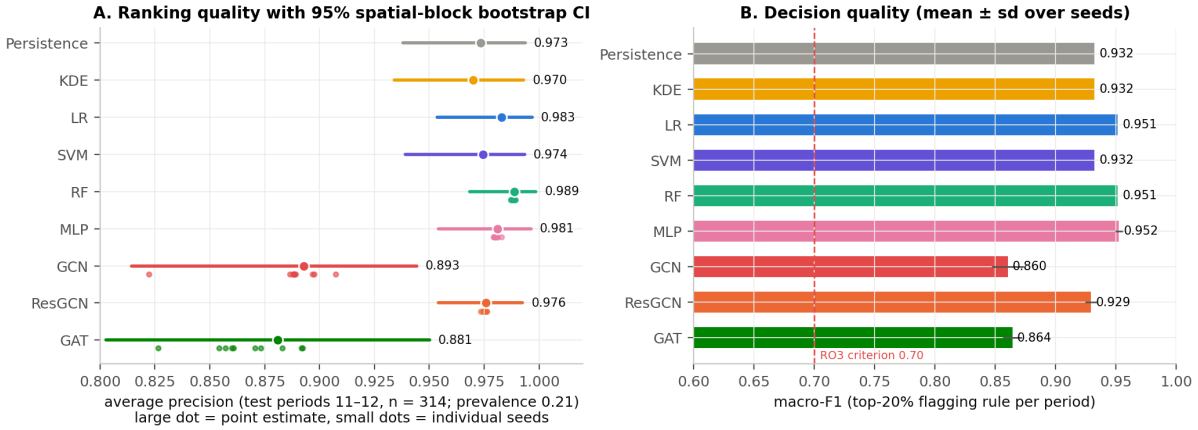


Figure 4.1: Phase 2 test comparison. (A) AP with 95% spatial-block bootstrap intervals; small dots show individual seeds. (B) Macro-F1 under the operational top-20% rule; the dashed line is the RO3 criterion.

LR was chosen as the Phase 2 classifier based on the pre-stated criterion because it had the highest validation AP (0.968) of all models. The test confusion matrix for the top-20% rule (Figure 4.3) for LR is 244 true negatives, 4 false positives, 6 false negatives, and 60 true positives; precision 0.938 and recall 0.909 for the high class. Since the rule selects exactly 32 cells per period but there are 32–35 cells that are marked high (ties), some false negatives are inherent to the problem, as even a perfect ranking would miss the ties past the 32nd cell.

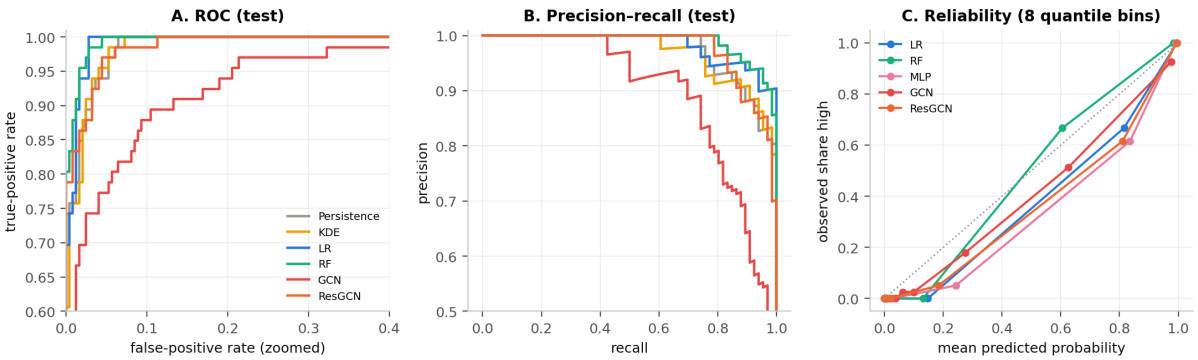


Figure 4.2: Phase 2 test ROC curves (zoomed), precision–recall curves and reliability diagrams. Persistence and KDE are ranking scores, so they have no reliability curve.

The curves in Figure 4.2 show the same ranking. The GCN's precision drops sharply beyond 50% recall, indicating that it confuses moderately high cells with their neighbours. The reliability diagram suggests that LR and the MLP are slightly overconfident in the 0.6–0.9 range—an expected effect of class weighting, which shifts probabilities upward—while RF is closest to the diagonal (Brier 0.023). Calibration is relevant to the application only when probabilities are shown; hence, the design shows ranks and calibrated bands instead of probabilities (Section 4.4).

4.1.4. Phase 1: Dominant Crime Type (E3)

Predicting the predominant category proved much harder (Table 4.4). The majority class baseline (BATTERY) scores a macro-F1 of 0.134; persistence scores 0.404; and LR scores 0.498, the largest test macro-F1. The GCN obtains 0.450 ± 0.019, and the GAT 0.450 ± 0.010, both lower than the SVM (0.467). The validation criterion favored the MLP (validation macro-F1 0.520 compared to 0.492 for LR), and the MLP scored 0.482 ± 0.018 in the test periods. The higher test result for LR is within the seed range and bootstrap confidence interval of the MLP, so no significant difference in performance can be concluded from this experiment. RQ1 is answered in the negative: the GCN does not perform better than the SVM or MLP.

Table 4.4: Phase 1 test results (periods 11–12; $n = 303$; support: battery 136, criminal damage 27, theft 111, other 29). Mean ± sd over 10 seeds where stochastic

Model	Accuracy	Macro-F1	Weighted F1	κ	F1 battery	F1 crim. damage	F1 theft	F1 other
Majority	0.366	0.134	0.196	0.000	0.000	0.000	0.536	0.000
Persistence	0.604	0.404	0.580	0.363	0.705	0.054	0.655	0.200
KDE	**0.640**	0.426	0.608	0.413	0.715	0.061	**0.711**	0.217
LR	0.611	**0.498**	**0.630**	0.425	**0.743**	**0.265**	0.660	0.324
SVM	0.564	0.467	0.597	0.388	0.717	0.192	0.615	0.344
RF	0.629 ± 0.007	0.460 ± 0.008	0.621	**0.437**	0.738	0.006	0.681	**0.418**
MLP (val. selected)	0.593 ± 0.013	0.482 ± 0.018	0.622	0.418	0.737	0.170	0.659	0.363
GCN	0.547 ± 0.016	0.450 ± 0.019	0.583	0.360	0.711	0.154	0.593	0.343
ResGCN	0.567 ± 0.021	0.467 ± 0.025	0.600	0.383	0.692	0.166	0.656	0.353
GAT	0.550 ± 0.012	0.450 ± 0.010	0.584	0.362	0.715	0.163	0.592	0.331

The per-class results indicate why this is difficult. BATTERY and THEFT are reasonably well predicted (F1 0.66 to 0.74) as there are a large number of cells where one type dominates consistently. However, CRIMINAL DAMAGE is the dominant type in just 27 test cell-periods,

and even when it is the dominant type it is usually only narrowly leading theft or battery, so even the best model gets an F1 of 0.27; RF effectively never predicts it. The confusion matrices (Figure 4.3) reveal that persistence tends to confuse BATTERY and THEFT the most (30 and 23 swaps), since both counts in mixed cells are similar. LR sacrifices some THEFT accuracy for more recall on the two minority classes, which explains why its accuracy is lower than that of KDE, but its macro-F1 is higher. The error map (Fig. 4.4B) indicates that Phase 1 errors are located mostly in the north-west and far south, where the cells have low volumes and mixed profiles, while the dense cells in the centre and south with a clear battery or theft profile are hardly ever wrong.

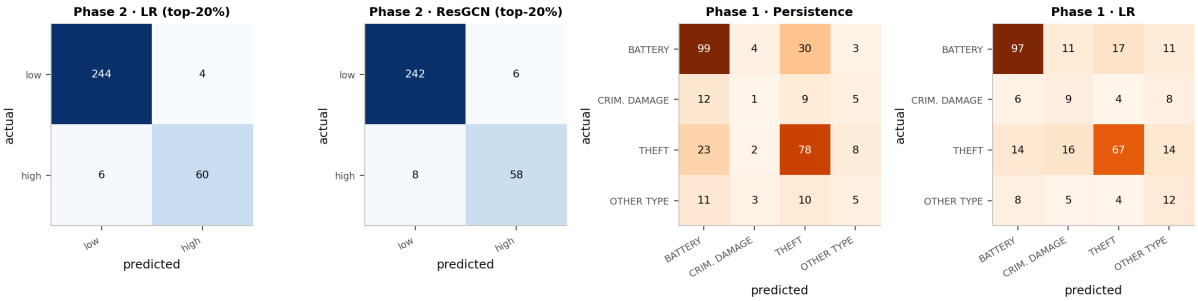


Figure 4.3: Test confusion matrices: Phase 2 under the top-20% rule (LR, ResGCN) and Phase 1 (persistence, LR).

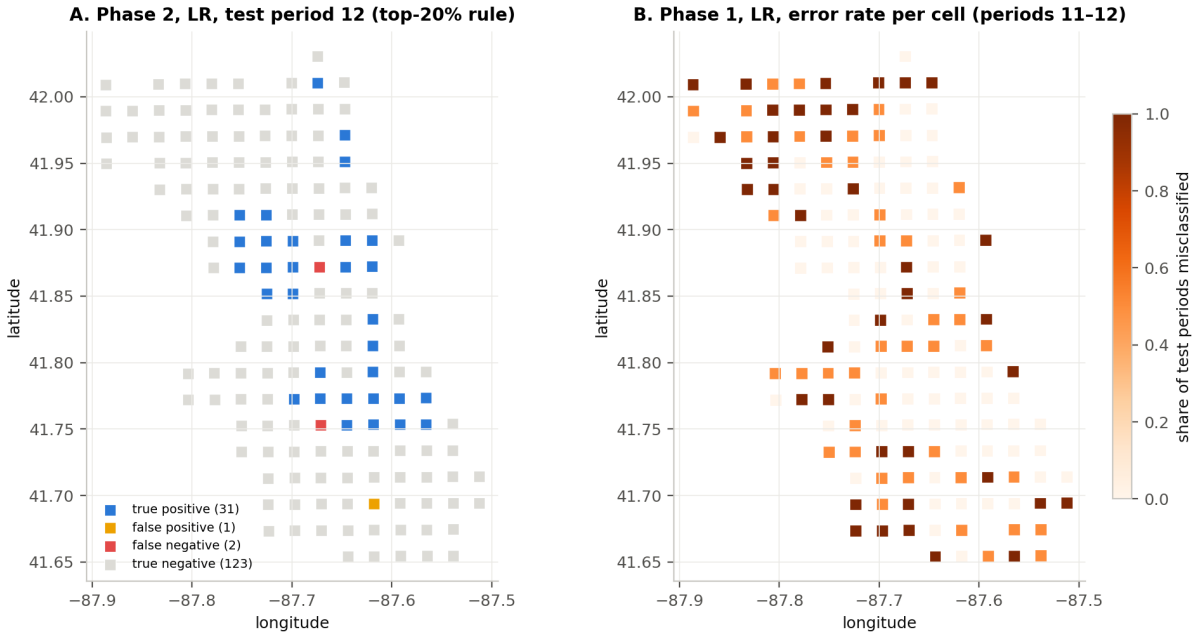


Figure 4.4: Geographic error analysis on test periods. (A) Phase 2 LR outcomes in period 12. (B) Share of the two test periods in which LR misclassified each cell's dominant type.

Phase 2 errors are rare and exist on the edge of the high category (Figure 4.4A). During period 12, the threshold was 71 incidents: one false negative (86 incidents), which was ranked 33rd, just outside the top 32 cells, and another, which was exactly equal to the threshold value of 71; the single false positive, a far-south cell with a high past volume, recorded 61 incidents and was ranked 31st. No geographic area is systematically missed, although evidence is available

for only two testing episodes.

4.1.5. Class-Imbalance Strategies (E4)

Table 4.5: Imbalance strategies for Phase 2 (test). Threshold metrics depend on the decision threshold; AP does not

Strategy	Threshold	AP	Precision	Recall	Pos. F1	Macro-F1
LR, no weighting	0.5	0.985	0.926	0.955	0.940	0.962
LR, no weighting	validation-tuned	0.985	0.868	1.000	0.930	0.955
LR, class weights	0.5	0.983	0.868	1.000	0.930	0.955
LR, class weights	validation-tuned	0.983	0.944	0.773	0.850	0.907
LR, SMOTE (train only)	0.5	0.985	0.892	1.000	0.943	0.963
LR, SMOTE (train only)	validation-tuned	0.985	0.948	0.833	0.887	0.930
GCN, unweighted CE	0.5	0.881	0.922	0.647	0.760	0.854
GCN, weighted CE	0.5	0.885	0.738	0.824	0.765	0.841
GCN, weighted CE	validation-tuned	0.885	0.870	0.717	0.784	0.866

There are three points to note about Table 4.5. First, none of the strategies modified the ranking quality, with LR's AP varying between 0.983 and 0.985, as reweighting mostly affects the intercept. Second, the strategies modified the precision-recall trade-off for a given threshold, such that class weighting and SMOTE gave LR perfect recall at the cost of precision, and class weighting shifted the GCN's recall from 0.647 to 0.824. Third, optimising the threshold on only two validation periods harmed all LR variants on the test periods (e.g., macro-F1 0.955 $\rightarrow$ 0.907 with class weights) because the F1-optimal threshold on 314 validation samples is noisy. In the three-year replication (948 validation samples), this depends on the validation sample size: the optimised threshold was either neutral or beneficial (unweighted LR 0.917 $\rightarrow$ 0.934; class-weighted LR 0.935 $\rightarrow$ 0.933; SMOTE 0.933 $\rightarrow$ 0.927; weighted GCN 0.859 $\rightarrow$ 0.862), and class weighting again affected recall (unweighted LR 0.828, weighted 0.932) but not AP (0.959 vs 0.958). Thus, the application will use a prevalence-matched top-k strategy that requires no parameter tuning and can be recalculated each period, and will take into account a tuned threshold only after about six closed validation periods have been completed. There were no gains from SMOTE over class weights in either study, so class weights were used.

4.1.6. Graph Ablations: When Does Message Passing Help? (E5)

Table 4.6: Graph ablations (test; mean ± sd over 10 seeds). Pilot-style GCN versus residual GCN

			Phase 2 AP		Phase 1 macro-F1	
Graph / depth	Edges	Mean degree	GCN	ResGCN	GCN	ResGCN
Rook 3.0 km, inverse distance (pilot)	272	3.5	0.885 ± 0.023	0.975 ± 0.001	0.450 ± 0.019	0.467 ± 0.025
Rook 3.0 km, unweighted	272	3.5	0.870 ± 0.029	0.974 ± 0.002	0.448 ± 0.011	0.464 ± 0.020
Queen 3.2 km, inverse distance	528	6.7	0.848 ± 0.024	0.973 ± 0.001	0.441 ± 0.012	0.461 ± 0.026
Radius 4.5 km, inverse distance	765	9.7	0.779 ± 0.018	0.975 ± 0.001	0.426 ± 0.020	0.460 ± 0.020
Radius 6.6 km, inverse distance	1,640	20.9	0.729 ± 0.010	0.976 ± 0.001	0.386 ± 0.014	0.450 ± 0.022
Depth 1 (rook)	272	3.5	0.893 ± 0.009	0.959 ± 0.021	0.453 ± 0.013	0.474 ± 0.024
Depth 2 (rook)	272	3.5	0.885 ± 0.023	0.975 ± 0.001	0.450 ± 0.019	0.467 ± 0.025
Depth 3 (rook)	272	3.5	0.872 ± 0.011	0.973 ± 0.003	0.429 ± 0.016	0.466 ± 0.017
MLP (no graph)	0	0	0.980 ± 0.001		0.482 ± 0.018	
LR without / with spatial-lag features	–	–	0.983 / 0.983		–	

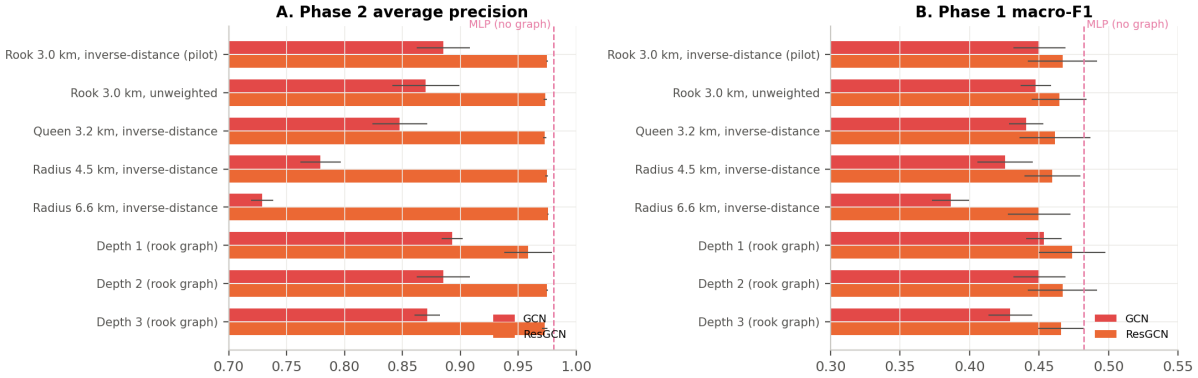


Figure 4.5: Graph ablations. The degradation in performance by the GCN increases as neighbourhood size and layer count increase, while the residual GCN becomes independent of the graph structure, approaching, but not exceeding, the MLP.

The ablation results (Table 4.6, Figure 4.5) support the smoothing hypothesis of Section 3.2.2. The AP of the baseline GCN model decreases monotonically from 0.885 to 0.729 with the increase of the mean degree from 3.5 to 20.9 and decreases with each additional layer, which is exactly what is predicted by the self-weight $1/(d_i + 1)$ and by the Laplacian-smoothing interpretation of Li et al. (2018). The elimination of the inverse distance weights made little difference (0.885 compared with 0.870 within one seed standard deviation), demonstrating that the uniform weights did not have any information content. The residual graph convolutional network, whose self connection is not weakened by degree, stayed flat on all of the graphs (0.973–0.976): it had learned to utilise each cell's past only and the neighbour connection contributed nothing measurable. Its pairwise difference from the MLP is -0.004 average precision (95% confidence interval $[-0.013, 0.008]$; Table 4.10). Adding neighbour-average features to LR also left AP unchanged (0.9830 vs 0.9834 AP). Across three independent tests, therefore, information from neighbouring cells did not improve forecasts once a cell's own history was known.

4.1.7. Feature Ablation and Leakage Check (E6)

Table 4.7: Feature-group ablation (test). GCN values are means over five seeds

		Phase 2 AP		Phase 1 macro-F1	
Feature set	Features	LR	GCN	LR	GCN
All features	23	0.983	0.896	0.498	0.448
Without victimisation history	17	0.974	0.895	0.502	0.443
Without crime-mix shares	15	0.983	0.932	0.493	0.457
Without coordinates	21	0.982	0.847	0.507	0.454
Victimisation lags only	4	**0.991**	0.867	0.340	0.391

Table 4.7 indicates that Phase 2 predictive performance comes almost entirely from victimisation history: four lagged features alone give LR an AP of 0.991, exceeding the full feature set (0.983). This suggests that the additional features may add a little variance without contributing further predictive signal. Removing victimisation history reduces LR's AP by only 0.009 because total-crime lags contain much of the same information. Phase 1 shows the opposite pattern: removing crime-mix shares barely changes LR's macro-F1 (0.493), whereas using victimisation lags alone reduces it to 0.340. This suggests that crime-type forecasting relies on the combined feature profile rather than any single feature group. Since all the features come before the target period, none of these results depends on leakage; the comparison with the pilot's perfect argmax rule (Table 4.2) reveals the extent to which a circular design would have exaggerated skill. The GCN benefits from excluding the composition shares (0.896 $\rightarrow$ 0.932), further evidence that aggregation mixes cell-level information with that of neighbours.

4.1.8. Robustness over Time, Space, Labels and Resolution (E7–E9)

Table 4.8: Rolling-origin evaluation: Phase 2 AP (top) and Phase 1 macro-F1 (bottom) by test period; each model is retrained on all earlier periods

Test period	8	9	10	11	12
Training target periods	3–6	3–7	3–8	3–9	3–10
P2 Persistence	0.938	0.965	**0.969**	0.970	0.977
P2 KDE	0.909	0.944	0.932	0.952	**0.991**
P2 LR	**0.944**	0.962	0.960	**0.977**	**0.991**
P2 GCN (5 seeds)	0.854	0.860	0.882	0.885	0.949
P2 ResGCN (5 seeds)	0.935	**0.965**	0.941	0.965	0.987
P1 Persistence	**0.458**	**0.446**	0.390	0.425	0.382
P1 LR	0.425	0.428	**0.567**	**0.490**	0.460
P1 GCN (5 seeds)	0.431	0.391	0.456	0.444	**0.472**

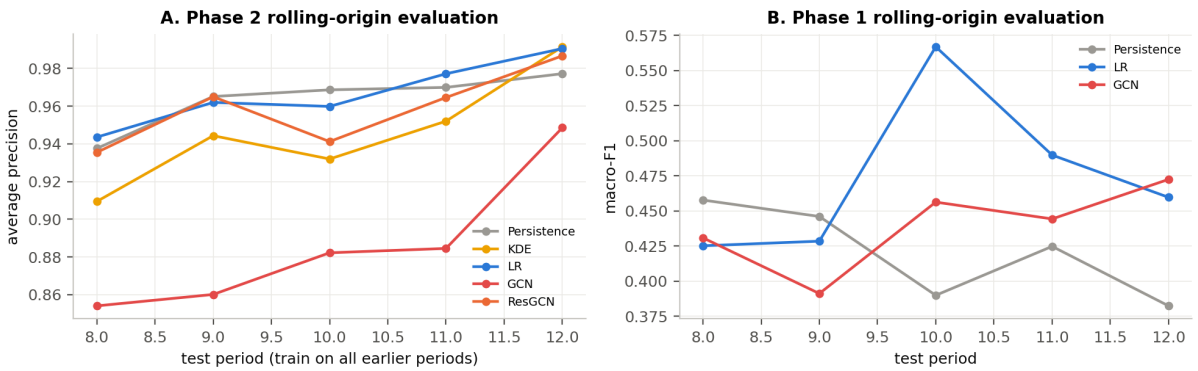


Figure 4.6: Rolling-origin evaluation across five test periods.

Over time. The rolling origin (Table 4.8, Figure 4.6) reveals that the Phase 2 ranking is stable: LR and persistence are within 0.01 AP of each other in four out of five windows, the GCN is the worst in all five windows, and all models get better as more training data becomes available and as summer volumes increase. LR outperforms persistence in three out of five windows, so its superiority is slight and inconsistent. The Phase 1 ranking is unstable: persistence outperforms in the first two windows (when LR has only four training periods), LR wins periods 10 and 11, while the GCN wins period 12. Conclusions drawn from Phase 1 should hence be considered tentative.

Table 4.9: Spatial-block cross-validation for Phase 2: held-out cells' labels masked during training; AP on held-out cells in test periods

Block	Cells	Test rows (high)	Persistence	LR	MLP	GCN	ResGCN
0	29	58 (10)	0.973	1.000	0.994	0.953	0.983
1	26	52 (10)	0.933	0.981	0.914	0.634	0.938
2	37	74 (13)	0.995	0.989	0.958	0.828	0.936
3	30	60 (2)	1.000	1.000	1.000	0.667	1.000
4	35	70 (31)	0.983	0.984	0.986	0.863	0.974
Mean			0.977	**0.991**	0.970	0.789	0.966

Over space. Holding out entire regions (Table 4.9) is the most difficult challenge for a graph model since the held-out cells' labels are unknown, but the features of their neighbours are known. LR transferred best (mean AP = 0.991), but the GCN performance declined sharply in blocks 1 and 3 (AP = 0.634 and 0.667). Block 3, located in the far south, contains only two high cell-periods, so its AP is highly unreliable and should not be over-interpreted. This indicates that the GCN's learning of smoothing does not transfer to regions with unfamiliar spatial structure, unlike history-based tabular models.

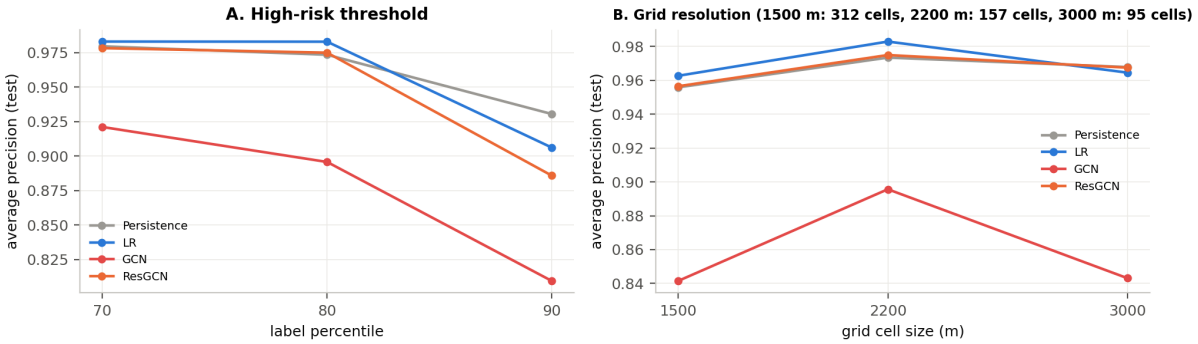


Figure 4.7: Sensitivity of Phase 2 test AP to the label percentile and to grid resolution.

Label threshold and grid size. Figure 4.7 and Appendix Table B.2 show that the conclusions are robust to the labelling choice: at the 70th percentile threshold (prevalence=0.30), all models apart from GCN have AP=0.978-0.983; at the 90th percentile (prevalence=0.10) AP is lowered for all models, with persistence (AP=0.930) ahead of logistic regression (AP=0.906), as the top decile represents a very stable population. When using cell sizes of 1.5 km (number of cells=312) and 3.0 km (number of cells=95), the ordering was maintained; namely, persistence, LR and ResGCN were within 0.01 of one another, with GCN lower by 0.09-0.13 AP relative to LR for each resolution tested. Finer grids result in a slight AP decline (0.956-0.963 at 1.5 km) because counts per cell are noisier.

4.1.9. Statistical Uncertainty and Paired Comparisons (E10)

Table 4.10: Spatial-block bootstrap (26 blocks of 3×3 cells, 2,000 resamples) for paired differences on the test periods

Comparison	Mean difference	95% CI	Share of resamples > 0
P2 AP: LR − Persistence	+0.009	$[-0.008, 0.031]$	0.86
P2 AP: RF − LR	+0.006	$[-0.002, 0.021]$	0.93
P2 AP: GCN − LR	−0.090	$[-0.152, -0.046]$	0.00
P2 AP: GCN − MLP	−0.089	$[-0.154, -0.046]$	0.00
P2 AP: ResGCN − GCN	+0.085	$[0.039, 0.154]$	1.00
P2 AP: ResGCN − MLP	−0.004	$[-0.013, 0.008]$	0.17
P2 AP: ResGCN − LR	−0.005	$[-0.018, 0.012]$	0.22
P2 AP: GAT − MLP	−0.098	$[-0.163, -0.039]$	0.00
P2 AP: GAT − GCN	−0.009	$[-0.059, 0.056]$	0.33
P1 macro-F1: LR − Persistence	+0.092	$[0.006, 0.167]$	0.98
P1 macro-F1: GCN − Persistence	+0.038	$[-0.037, 0.103]$	0.85

Table 4.10 provides the answer to RQ3. Resampling blocks of 6.6 km instead of individual cell-periods takes into account spatial correlation; in the case of the GCN, the confidence interval for AP ($[0.814, 0.944]$) is wider than under the naive independent bootstrap ($[0.831, 0.944]$), as expected when neighbouring errors are correlated. Three differences are clearly real: the GCN is inferior to the LR and the MLP models by about 0.09 AP, the residual GCN is superior

to the GCN by 0.085, the GAT is inferior to the MLP model by about 0.10 AP, and the superiority of LR over persistence in terms of crime type (+0.09 macro-F1, interval $[0.006, 0.167]$) is marginal and not confirmed by the permutation test (raw $p = 0.077$, Table 4.14). Three are not distinguishable from zero: LR vs. Persistence for victimisation, RF vs. LR, and ResGCN vs. MLP. The bootstrap accounts for uncertainty due to which neighbourhoods happened to be tested; seed uncertainty is given in Tables 4.3–4.4, and temporal uncertainty by the rolling origin. Because the intervals rely on only two test periods, they still understate uncertainty about future seasons. As intervals alone cannot form a test, E16 adds paired spatial-block permutation tests with Holm adjustment (Table 4.14), so effect sizes with intervals are treated as the principal evidence since the 26 blocks provide the test with limited power and lack of significance does not indicate equivalence.

4.1.10. Continuous Count Forecasting (E11)

Table 4.11: Next-period victimisation count forecasts (test; $n = 314$)

Model	MAE	RMSE	R^2	Spearman ρ
Persistence (last period)	6.59	9.54	0.939	0.973
Persistence (3-period mean)	5.99	**8.62**	**0.950**	**0.979**
Poisson GLM ($\alpha = 1$)	**5.95**	8.94	0.946	**0.979**
MLP (Poisson loss, 5 seeds)	6.70	9.98	0.933	0.977
GCN (Poisson loss, 5 seeds)	11.55	16.80	0.810	0.912

The continuous task promised in Assessment 1 validates the classification results (Table 4.11). A three-period moving average accounts for 95% of the variance in next-period counts and ties with, but does not outperform, a Poisson GLM. The GCN using a Poisson loss function performs significantly worse. Since the neural count models used the same hyperparameters as the classifiers and the GLM α was on the border of its grid, both were investigated later in E18 (Section 4.1.13), with tuned neural models and a larger α grid, and the conclusion did not change.

4.1.11. Unsupervised Hotspot Detection with DBSCAN (E12)

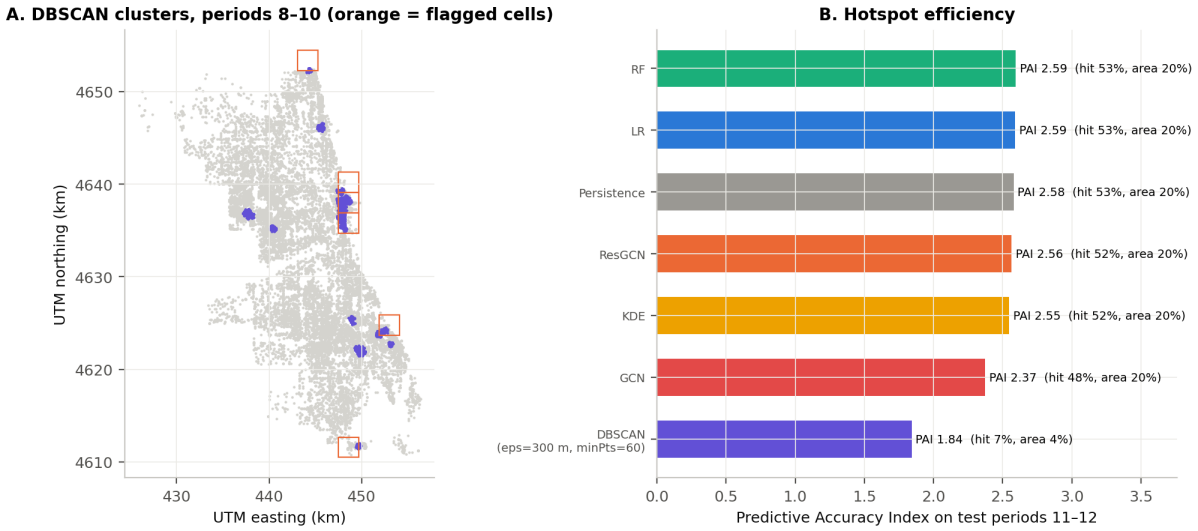


Figure 4.8: (A) DBSCAN clusters of victimisation incidents in periods 8–10 with flagged cells. (B) PAI on the test periods for DBSCAN zones and for the top-20% cells of each ranking model.

It is unclear whether KDE itself would satisfy the unsupervised criterion in the way defined by the lecturer, and this is flagged for confirmation: KDE uses no labels, but here it served as a forecasting baseline and was scored against labels. To include a truly unsupervised element (LO3), DBSCAN (Ester et al., 1996) was performed on the location data of victimisation incidents. The values of ε from the k-distance plot (30th neighbour) were used to create a grid ranging from 100 m to 300 m with minPts $\in \{30, 60, 100\}$; cells were marked as clusters when at least half of their incidents were clustered. Settings were chosen by PAI when fitted on periods 5–7 and validated on periods 9–10, re-fitted on periods 8–10 and tested on periods 11–12. The selected setting ($\varepsilon = 300$ m, minPts $= 60$) discovered 12 dense clusters (silhouette 0.63 among clustered points), marked 90% of incidents as noise, identified six cells (3.8% of the area) and captured 7.0% of victimisation incidents from the test periods: a PAI value of 1.84 (Figure 4.8). The supervised rankings with top-20% cells led to PAI of 2.37–2.59, capturing around 53% of incidents in 20% of cells. A more liberal setting ($\varepsilon = 300$ m, minPts $= 30$) resulted in 36 flagged cells and PAI of 2.10 during the test period, but was not selected during the validation. The overlap between flagged cells in two distinct three-period windows was moderate (Jaccard 0.50).

These results highlight the strengths and limitations of the unsupervised approach. DBSCAN detects compact micro-hotspots within cells, which can be used as an informative layer to indicate *within the flagged cell* where incidents cluster, but it is parameter dependent, and PAI gives high scores to very small areas; thus the validation-selected setting proved too conservative for area-level planning. The other density-based approach, KDE, achieved a much higher PAI (2.55). Thus, in this application, DBSCAN is a descriptive layer, not a decision-making tool.

4.1.12. Computational Cost (E13)

Table 4.12: Measured computational cost (Apple M5 CPU; Phase 2; mean over seeds)

Model	Trainable parameters	Fit time (s)	Inference for 314 cells (ms)	Best epoch
LR	24	0.004	0.2	–
SVM (RBF)	support vectors	0.003	1.7	–
RF (300 trees)	–	0.21	25	–
MLP	3,330	0.19	0.2	160
GCN	1,666	0.45	0.4	226
ResGCN	6,660	0.77	0.5	278
KDE (per period, 18k points)	–	–	38	–

All models train in under one second on a laptop CPU and evaluate the entire city in milliseconds (Table 4.12), and the full experiment set with 14 experiment groups and hundreds of model fits took under ten minutes, with peak memory use under 1 GB. Thus computational cost is not an issue at this level of analysis. It becomes an issue only at much higher resolutions or with daily retraining, where the GCN's block-diagonal snapshot graph scales linearly with periods $\times$ cells.

4.1.13. Post-hoc Extensions: Preprocessing, Significance, Features, Counts and Effective Sample Size (E15–E19)

These five analyses were added after the main test results became available to fill in any gaps identified while the report was evaluated using the rubric and Assessment 1. In order to avoid selection bias, each one of them uses validation data exclusively, without modifying the models selected in Sections 4.1.3–4.1.4. If one of them contradicts a previous statement, then that statement has been corrected.

Preprocessing (E15). Table 4.13 assesses how much each preprocessing step helps. Validation favours the full preprocessing pipeline for both LR models (validation AP 0.968 compared to 0.959 without; validation macro-F1 0.492 compared to 0.454), and the robust scaler performs the same. In test time frames, raw features provided a better AP for LR (0.993 compared to 0.983, with the penalty C tuned to 0.01), an advantage too small for the bootstrap to resolve (about 0.01) and one that validation did not predict; using it would be selecting based on the test data, so it was not used. For Phase 1, the log transform and scaling clearly help (LR 0.498 vs 0.446 raw, MLP 0.477 vs 0.421). In the case of the GCN model, scaling matters, not the log transform (log transform 0.835, z-score 0.904, both 0.896), since the propagation technique relies heavily on the mixing of features across nodes, and unscaled counts dominate. Omitting the 17 low-volume cells from training did not benefit the logistic regression model (0.982) but hurt the GCN (0.781 versus 0.896) since the GCN requires every node to learn its neighbourhoods. The supported claim is thus a modest one, namely that preprocessing is essential to the neural network and Phase 1 models, retained on validation and data-quality grounds but does not explain the model rankings.

Table 4.13: Preprocessing ablation (E15, post-hoc). LR is re-tuned per variant on validation; MLP, ResGCN and GCN (5 seeds) keep their E1/E3 hyperparameters

	Phase 2 AP					Phase 1 macro-F1		
Variant	LR val.	LR test	MLP	ResGCN	GCN	LR val.	LR test	MLP
log + z-score (main)	0.968	0.983	0.980	0.975	0.896	0.492	0.498	0.477
raw	0.959	0.993	0.983	0.961	0.875	0.454	0.446	0.421
z-score only	0.960	0.982	0.977	0.974	0.904	0.473	0.476	0.462
log only	0.958	0.985	0.979	0.969	0.835	0.486	0.482	0.410
log + robust scaling	0.970	0.982	0.981	0.974	0.888	0.490	0.498	0.463
Drop 17 low-volume cells (train)	–	0.982	0.973	0.973	0.781	–	–	–

Significance tests (E16). Table 4.14 provides paired spatial-block permutation tests. The two model scores (percentile ranks within periods for Phase 2; predictions for Phase 1) are permuted within randomly selected 3×3-cell blocks and the AP or macro-F1 difference is recalculated 4,000 times, and p is the proportion of permutations with a difference of at least the observed magnitude; Holm's correction is done within each family. In the one-year analysis three differences remain after correction: GCN is inferior to MLP ($p_{\text{Holm}} = 0.010$), residual GCN is superior to GCN (0.010), and GAT is inferior to MLP (0.032). None of the comparisons between LR, RF, MLP, SVM, ResGCN and persistence for victimisation is significant (lowest raw $p = 0.18$), consistent with the bootstrap. For crime type, no difference remains after correction; LR vs. persistence has raw $p = 0.077$ (Holm 0.53), which means that the improvement observed in Section 4.1.4 is only tentative. The three-year columns tell the same story.

Table 4.14: Paired spatial-block permutation tests (E16, post-hoc; 26 blocks, 4,000 permutations; Holm-adjusted within each task)

	One-year study			Three-year study		
Comparison (difference in AP or macro-F1)	Diff.	p	p_{Holm}	Diff.	p	p_{Holm}
P2 AP: LR − Persistence	+0.009	0.336	1.000	+0.002	0.612	1.000
P2 AP: RF − LR	+0.005	0.248	1.000	+0.006	0.074	0.518
P2 AP: MLP − LR	-0.001	0.797	1.000	+0.003	0.543	1.000
P2 AP: SVM − LR	-0.008	0.176	1.000	-0.005	0.335	1.000
P2 AP: GCN − MLP	-0.088	0.001	0.010	-0.082	0.000	0.002
P2 AP: ResGCN − MLP	-0.006	0.352	1.000	+0.004	0.291	1.000
P2 AP: ResGCN − GCN	+0.082	0.001	0.010	+0.086	0.000	0.002
P2 AP: GAT − MLP	-0.100	0.004	0.032	-0.089	0.002	0.018
P2 AP: GAT − GCN	-0.012	0.695	1.000	-0.007	0.781	1.000
P2 AP: KDE − Persistence	-0.002	0.747	1.000	-0.013	0.293	1.000
P1 F1: LR − Persistence	+0.095	0.076	0.527	+0.043	0.138	0.589
P1 F1: GCN − SVM	-0.019	0.419	1.000	-0.052	0.059	0.357
P1 F1: GCN − MLP	-0.032	0.202	1.000	-0.040	0.118	0.589
P1 F1: LR − GCN	+0.050	0.075	0.527	+0.030	0.245	0.736
P1 F1: ResGCN − GCN	+0.029	0.224	1.000	+0.046	0.011	0.079
P1 F1: GAT − GCN	+0.004	0.822	1.000	+0.004	0.698	0.810
P1 F1: MLP − LR	-0.018	0.559	1.000	+0.010	0.405	0.810

Phase 1 feature extension (E17). Eighteen additional history-only features were introduced (previous period and 6-period group ratios; dominant class features for final period and last three periods; dominance margin and mix entropy). The validation macro-F1 in Table 4.15 improved for RF (0.486 → 0.518) and ResGCN (0.490 → 0.514), was unchanged for MLP but declined for LR (0.492 → 0.466). Hence selection based on validation leads to extended features for RF, MLP and ResGCN but base features for LR. In the testing periods, the extended RF model performed better (0.461 → 0.518), whereas the extended MLP and ResGCN models performed poorly (0.477 → 0.434 and 0.462 → 0.433). Given about 300 test rows and eight model-feature combinations, the best result (extended-feature RF, 0.518) is optimistic. No feature set resolves the weakness of Phase 1.

Table 4.15: Phase 1 feature extension (E17, post-hoc): macro-F1 on validation and test periods

	Base (23 features)		Extended (+18)		Chosen on
Model	Validation	Test	Validation	Test	validation
LR	0.492	0.498	0.466	0.496	base
RF	0.486	0.461	0.518	0.518	extended
MLP	0.520	0.477 ± 0.020	0.521	0.434 ± 0.032	extended
ResGCN	0.490	0.462 ± 0.026	0.514	0.433 ± 0.014	extended

Tuned count models (E18). Table 4.16 repeats the count comparison with neural models tuned on validation (24 configurations × 3 seeds each) and an extended Poisson α grid up to 100; the selected $\alpha = 1$ is now interior. Tuning improved the MLP (MAE 6.27 vs 6.70) but it remains behind the three-period moving average (5.99) and the GLM (5.94). The tuned GCN is still markedly worse (MAE 11.47, $R^2 = 0.81$), the ResGCN is mid-table (7.24) and the GAT is worst (16.5, $R^2 = 0.55$). Missing tuning is therefore not the explanation for the GCN's count failure, and the smoothing explanation stands.

Table 4.16: Tuned next-period victimisation count forecasts (E18, post-hoc; test periods, $n = 314$)

Model	MAE	RMSE	R^2	Spearman ρ
Persistence (last period)	6.59	9.54	0.939	0.973
Persistence (3-period mean)	5.99	8.62	0.950	0.979
Poisson GLM (tuned alpha)	5.94	8.94	0.946	0.979
MLP (Poisson loss, tuned)	6.27	9.37	0.941	0.977
GCN (Poisson loss, tuned)	11.47	16.89	0.808	0.912
ResGCN (Poisson loss, tuned)	7.24	10.92	0.920	0.972
GAT (Poisson loss, tuned)	16.51	25.78	0.552	0.865

Effective sample size and a weighted graph (E19). Table 4.17 shows just how much independent information the test sets actually contain. Moran's I statistic of log victimisation counts is around 0.5, and the lag-1 autocorrelation of the periods is around 0.96. Applying the heuristic $n_{\text{eff}} \approx n(1-\rho)/(1+\rho)$ to each of the dependencies, the 314 test rows of the main study correspond to only 47 effectively independent observations (roughly the number of cells), and the 948 rows from the three-year study to about 50, since the additional time periods are almost

entirely redundant. It is an approximation rather than an accurate count, but it explains why differences of about 0.01 AP cannot be resolved and why the number of cells, not incidents or rows, limits the evidence. The table also provides the information on the centroid-based graph (distances between incident centroids from periods before the test window, according to the base paper): it is denser (mean degree 4.9-5.0) and its edge length varies (0.68-3.0 km), so the inverse distance weight does matter. Still, the GCN reached only 0.770 and the GAT 0.800 (one year), compared with 0.973 for the ResGCN, which means that the constant weight in the lattice did not cause the GCN's poor performance.

Table 4.17: Effective sample size and centroid-graph check (E19, post-hoc)

Quantity	One-year study	Three-year study
Cells (nodes)	157	158
Test rows (cells × periods)	314	948
Moran's I of log victimisation counts (mean over test periods)	0.540	0.519
Lag-1 correlation of log counts between periods	0.964	0.965
Effective spatial sample $n(1-I)/(1+I)$	46.7	50.1
Effective independent periods per cell (floored at 1)	1.0	1.0
Approximate effective sample size of the test set	46.7	50.1
Centroid graph: undirected edges	386	396
Centroid graph: mean degree	4.92	5.01
GCN on centroid graph: AP / Phase 1 macro-F1	0.770 / 0.438	0.852 / 0.428
ResGCN on centroid graph: AP / Phase 1 macro-F1	0.973 / 0.469	0.962 / 0.464
GAT on centroid graph: AP / Phase 1 macro-F1	0.800 / 0.434	0.849 / 0.424

4.1.14. Three-Year Replication (E21)

The primary experiment spans one year divided into two test periods. To establish whether the results of this study are an artefact of this time span, the entire comparison was reproduced over three years, from 24 August 2023 through 23 August 2026. The first two of these three years (509,905 records) are sourced from the same portal's *Crimes – 2001 to Present* database and concatenated with the one-year sample, which totals 739,180 records, 732,017 of which are cleaned. There are 158 occupied cells and 39 full 28-day periods in this concatenated sample. Period counts and victimisation, arrest and domestic shares show no discontinuity at the point of concatenation (period 26, covering it, contains 18,659 records, compared to 19,521 and 19,061 for adjacent periods). Protocol remains the same: features rely on past periods only; training data comprises periods 3–26 (up to 17 September 2025), validation periods 27–32 and test periods 33–38 (5 March to 19 August 2026); hyperparameters were tuned using validation only; the Phase 1 classes were derived again from training periods using the 5% criterion and retained the classes of battery, criminal damage and theft (shares 33%, 6%, 52%). Each model was executed with ten seeds.

Table 4.18 and Figure 4.9 confirm the replicability of the results. In Phase 2, RF (AP 0.965), the residual GCN (0.964), the MLP (0.958), LR (0.958), the SVM (0.952) and persistence (0.951) constitute one group where there is no statistically significant difference (LR minus persistence +0.007, 95% CI $[-0.005, 0.016]$; RF minus LR $p = 0.074$, Holm 0.52). The GCN (0.875) and

GAT (0.870) are definitely poorer (GCN minus MLP −0.084, $p_{\text{Holm}} = 0.003$), so the quadrupling of the training periods from 6 to 24 did not close the graph gap. For Phase 1, RF (macro-F1 = 0.499) and SVM (0.492) take first place while GCN (0.443) ranks last among the learned models, and the benefit of LR over persistence decreases to only +0.043 ($p = 0.14$). The validation procedure would now select RF to proceed to Phase 2 (validation AP = 0.966 compared with 0.963 for LR), which is consistent with RF's test ranking and still within noise of LR.

Table 4.18: Three-year replication (E21): test periods 33–38 ($n = 948$ cell-periods, 192 high; Phase 1 support 899). Mean ± sd over 10 seeds; CI from the spatial-block bootstrap

	Phase 2 (three years)			Phase 1	Phase 2 AP
Model	AP	AP 95% CI	Top-20% macro-F1	macro-F1	one-year study
Persistence	0.951	[0.899, 0.979]	0.935	0.427	0.973
KDE	0.943	[0.875, 0.978]	0.925	0.467	0.970
LR	0.958	[0.904, 0.984]	0.928	0.469	0.983
SVM	0.952	[0.889, 0.980]	0.915	0.492	0.974
RF	0.965 ± 0.000	[0.923, 0.988]	0.932 ± 0.001	0.499 ± 0.004	0.988
MLP	0.958 ± 0.002	[0.907, 0.985]	0.931 ± 0.003	0.475 ± 0.008	0.980
GCN	0.875 ± 0.006	[0.780, 0.930]	0.865 ± 0.004	0.443 ± 0.006	0.885
ResGCN	0.964 ± 0.001	[0.923, 0.987]	0.929 ± 0.003	0.474 ± 0.014	0.975
GAT	0.870 ± 0.007	[0.788, 0.932]	0.862 ± 0.006	0.442 ± 0.009	0.867

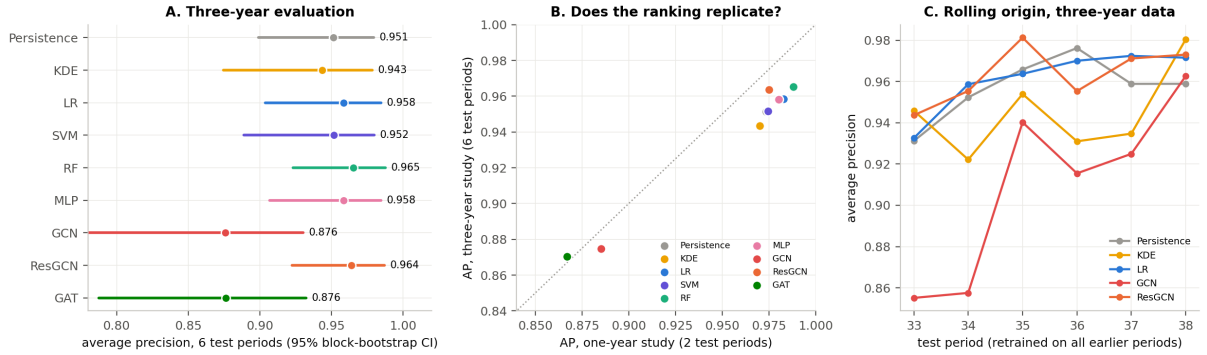


Figure 4.9: Three-year replication. (A) AP with 95% block-bootstrap intervals. (B) Model-by-model AP in the one-year and three-year studies, close to the diagonal for every model. (C) Rolling-origin AP over the test periods.

The long rolling origin includes the seasons from the second year (12 windows, test periods 15–37; retraining on all previous periods in each window). Table 4.19 confirms that the GCN is worse than persistence, LR, and the residual GCN for all 12 windows, that LR is better than persistence for only 6 of 12 windows (mean AP 0.966 versus 0.968), and that the residual GCN equals persistence on average (0.968). For crime type, LR is better than persistence for 7 of 12 windows (mean macro-F1 0.494 versus 0.470), a difference that is consistent in sign but small. Holding out spatial blocks (E9) gives mean AP 0.897 for LR, 0.890 for the residual GCN, 0.886 for MLP, 0.884 for persistence and 0.742 for the GCN; the GCN is better only in block 0, which has only six high cell-periods out of 180, too few to interpret. In addition, the data for three years provide a second, independent check on the earlier findings about decision rules (Section 4.1.5), imbalance and the graph (Table 4.17).

Table 4.19: Long rolling origin on three-year data (E21): Phase 2 AP and Phase 1 macro-F1 by test period (start date); 5 seeds for the GCN and ResGCN

	Phase 2 AP					Phase 1 macro-F1		
Test period	Persistence	KDE	LR	GCN	ResGCN	Persistence	LR	GCN
15 (17 Oct 2024)	0.987	0.979	0.972	0.919	0.974	0.490	0.516	0.478
17 (12 Dec 2024)	0.989	0.977	0.992	0.951	0.988	0.538	0.487	0.458
19 (6 Feb 2025)	0.981	0.949	0.973	0.896	0.962	0.468	0.515	0.479
21 (3 Apr 2025)	0.969	0.971	0.951	0.894	0.969	0.510	0.504	0.457
23 (29 May 2025)	0.960	0.963	0.954	0.903	0.950	0.480	0.531	0.517
25 (24 Jul 2025)	0.988	0.956	0.994	0.950	0.991	0.446	0.414	0.381
27 (18 Sep 2025)	0.954	0.970	0.957	0.898	0.950	0.389	0.527	0.505
29 (13 Nov 2025)	0.972	0.949	0.981	0.926	0.982	0.527	0.516	0.482
31 (8 Jan 2026)	0.960	0.935	0.950	0.898	0.957	0.415	0.461	0.438
33 (5 Mar 2026)	0.931	0.946	0.933	0.855	0.944	0.506	0.500	0.453
35 (30 Apr 2026)	0.966	0.954	0.964	0.940	0.981	0.479	0.489	0.388
37 (25 Jun 2026)	0.959	0.935	0.972	0.925	0.971	0.386	0.463	0.454
Mean	0.968	0.957	0.966	0.913	0.968	0.470	0.494	0.457

4.1.15. Extension Experiments: Label Definition, Error Disparity and Boosting (E22–E24)

Three gaps that remained after the first full draft were closed with further post-hoc experiments. As with E15–E19, every choice in them used validation periods only, and none of them changed the selected models.

Narrower victimisation label (E22). The main label counts 11 person-directed categories, which is broad (Appendix A). E22 re-ran the Phase 2 comparison with a much narrower label: FBI index violent crime only (codes 01A homicide, 02 criminal sexual assault, 03 robbery, 04A aggravated assault and 04B aggravated battery), which covers 20,376 incidents (8.9% of cleaned records, against 32.2% for the main label). All features and the top-20% label were rebuilt from this definition, and every model was re-tuned on periods 9–10. The two labels agree on 93.6% of test cell-periods (Jaccard overlap of the high cells 0.75). Table 4.20 shows that the ranking of methods is unchanged: persistence (AP 0.943), LR (0.940), KDE (0.939), RF (0.932) and MLP (0.931) are again within about 0.01 of one another, the residual GCN is close behind (0.920), and the GCN is again last (0.874). All AP values are about 0.04 lower than for the broad label because rarer events give noisier counts. The conclusions in Section 4.1.3 therefore do not depend on how broadly victimisation is defined.

Table 4.20: Phase 2 test results with the narrower FBI index violent-crime label (E22; periods 11–12; 5 seeds for RF and neural models)

Model	AP	ROC-AUC	Top-20% precision	Top-20% recall	Top-20% macro-F1
Persistence	0.943	0.979	0.891	0.792	0.897
LR	0.940	0.979	0.906	0.806	0.906
KDE	0.939	0.978	0.906	0.806	0.906
RF	0.932	0.979	0.900	0.800	0.902
MLP	0.931	0.976	0.900	0.800	0.902
ResGCN	0.920	0.973	0.891	0.792	0.897
GCN	0.874	0.950	0.819	0.728	0.854

Error disparity across areas (E23). The earlier draft could not examine whether errors fall unevenly on poorer areas because no area-level socio-economic data had been joined. E23 joined the City of Chicago's community-area boundaries (City of Chicago, 2026a) and its 2008–2012 census indicators (City of Chicago, 2026b) to the incidents (99.7% of incidents fell inside a community area), gave each cell the incident-weighted mean of the hardship index of its community areas, and split the 156 matched cells into three equal hardship terciles (52 cells, 104 test cell-periods each). Test predictions from E2 and E3 were then compared across terciles, with 95% intervals from a bootstrap that resamples cells (both test periods of a cell kept together). Table 4.21 gives the results for the selected LR model, persistence and the GCN.

Table 4.21: Test errors by community-area hardship tercile (E23; 104 test cell-periods per tercile; 95% cell-bootstrap intervals in brackets)

		Phase 2 (top-20% rule)				Phase 1	
Model	Hardship tercile	Share high (true)	Share flagged	Recall	False-positive rate	Error rate	Macro-F1
LR	Low	0.125	0.125	0.85 [0.60, 1.00]	0.022 [0.00, 0.06]	0.49	0.34
	Middle	0.163	0.154	0.82 [0.58, 1.00]	0.023 [0.00, 0.06]	0.44	0.46
	High	0.346	0.337	0.97 [0.90, 1.00]	0.000 [0.00, 0.00]	0.24	0.56
Persistence	Low	0.125	0.106	0.69 [0.33, 0.92]	0.022 [0.00, 0.06]	0.39	0.40
	Middle	0.163	0.173	0.94 [0.78, 1.00]	0.023 [0.00, 0.06]	0.44	0.36
	High	0.346	0.337	0.92 [0.78, 1.00]	0.029 [0.00, 0.07]	0.36	0.32
GCN	Low	0.125	0.106	0.54 [0.15, 0.90]	0.044 [0.00, 0.10]	0.52	0.29
	Middle	0.163	0.144	0.53 [0.19, 0.83]	0.069 [0.01, 0.13]	0.48	0.44
	High	0.346	0.365	0.97 [0.91, 1.00]	0.044 [0.00, 0.12]	0.35	0.41

Three findings follow. First, the *selection rate* is strongly unequal: LR flags 33.7% of cell-periods in high-hardship areas against 12.5% in low-hardship areas. This mirrors the recorded base rates almost exactly (34.6% vs 12.5%; Spearman correlation between hardship and test victimisation counts 0.37), so the model is not adding a disparity of its own, but it does mean that any use of the ranking directs attention mainly to poorer areas, which is the setting in which the stigma and feedback-loop risks of Section 4.3 apply. Second, *error rates* for Phase 2 are close to parity: recall and false-positive rates for LR overlap across terciles, and the only visible gap (lower recall in low-hardship areas, 0.85 vs 0.97) has wide intervals because each tercile holds only 13–36 high cell-periods. Where there is a difference, high victimisation in better-off areas is the more likely to be missed, so these areas are not over-flagged. Third, Phase 1 is *clearly less accurate in low-hardship areas* (LR error rate 0.49 vs 0.24; macro-F1 0.34 vs 0.56), due to their

heterogeneous crime profile and low volumes (Figure 4.4B). This supports displaying the type-of-crime prediction as only low-reliability context. This audit is restricted to the point where the census indicators are from 2008–2012; it is comparing neighborhoods, it uses labels that are based on recorded crime and thus suffer from the same reporting problem, and the intervals are large. Therefore, it is just an error-disparity audit, not a comprehensive group-fairness audit.

Gradient boosting (E24). To determine whether a stronger tabular learner would have an effect on model selection, histogram gradient boosting (HGB; class-weighted, L2 penalty 1.0) was tuned on the validation periods over learning rate $\{0.03, 0.1\}$, depth $\{3, \text{unlimited}\}$ and $\{100, 300\}$ iterations. In Phase 2, its best validation AP (0.962) was lower than LR's (0.968), so that the aforementioned rule would not select it; on test, it scored AP 0.975 and top-20% macro-F1 0.922 (5 seeds), in-between persistence and LR. In Phase 1, HGB scored macro-F1 0.463 (accuracy 0.584, $\kappa = 0.37$; validation 0.526), lower than LR (0.498) and suffering from the same deficiency in predicting CRIMINAL DAMAGE (F1 0.20). Thus, a more flexible learner does not add any value here, which is consistent with E17 and E19: the problem is not the model capacity but the few independent cells.

4.2. Discussion

RQ1 (crime type). No. Under a leakage-safe forecasting design the GCN (macro-F1 0.450) did not outperform the SVM (0.467), the MLP (0.482) or LR (0.498), and the GAT performed as well as the GCN (0.450). The pilot's apparent superiority (66.67% vs 37.50% accuracy) was due to a circular problem. Learned models have not been found to outperform persistence for crime type: LR's +0.09 macro-F1 (bootstrap CI $[0.006, 0.167]$) is borderline, and the permutation test yields $p = 0.077$ (Holm 0.53). In the three-year replication the LR's advantage is reduced to +0.04 ($p = 0.14$), RF (0.499) and the SVM (0.492) perform best, and the GCN (0.443) is again inferior to them; additional history variables (E17) improved RF (0.518) but not the neural models. RO2 is thus partially achieved.

RQ2 (victimisation). Yes for history-based features; no for the graph. LR selects the top-20% cells with precision 0.938 and recall 0.909 (macro-F1 0.951), comfortably meeting the RO3 threshold, and thus RO3 is fulfilled. The graph did not bring value in any of the three independent evaluations (graph vs MLP, neighbour features in LR, spatial block transfer). This is contrary to the prediction in Assessment 1 and Zubair et al. (2025), but is consistent with the theory, as the target here is the cell's own rank, the rankings are persistent with $\rho \approx 0.97$, and symmetric normalisation removes precisely this type of information. It also confirms the high persistence baselines found in the hotspot literature (Bowers et al., 2004; Chainey et al., 2008).

RQ3 (signal versus chance). The predictive skill is real and mainly attributable to persistence. All models are significantly better than the nulls (AP 0.21; macro-F1 0.44); the order of the models is stable in five test windows, three grid resolutions, a three-year replication study, and a 12-window rolling-origin analysis; and the GCN's deficit relative to the other models is

significant ($p_{\text{Holm}} = 0.010$ one-year, 0.003 three-year). The *increment* in learned models over persistence on victimisation (of around 0.01 AP) is within the margin of error (LR - persistence $p = 0.34$; LR beats persistence in only 6 out of 12 rolling windows). Since the test sets have only about 47-50 independent observations (E19), such a difference is not resolvable. The 4.4-fold difference in future victimisation between high- and low-risk cells in the proposal is a true description, but it is due to persistence of concentration rather than to model sophistication.

Model selection. Using the pre-stated rule, **LR is chosen as the Phase 2 model** in the main analysis: the highest validation AP, tied with the best test AP (RF, +0.006 AP, CI includes zero), the best generalisation to new regions, 24 interpretable parameters, and millisecond training time. In the three-year replication, validation favours RF (0.966 vs 0.963), but still within noise, and persistence is a valid zero-cost alternative; LR is recommended on transparency grounds, while RF and ResGCN are monitored challengers. For Phase 1, the validation rule chose the MLP network; however, since LR is statistically equivalent and interpretable, the application displays only the crime type prediction of the LR as *contextual information*. The main recommendation — prefer an interpretable model over a more complex one, when the latter does not bring anything quantifiable — is derived from the results rather than from the original preference for GCNs.

Unexpected findings. Four findings were unexpected. (i) LR reproduced the pilot result exactly. (ii) GCN performance declined monotonically as graph size increased, making graph size a diagnostic for over-smoothing. Neither the GAT with attention learning nor the weighted centroid graph approach could solve the problem. (iii) The impact of threshold adjustment was influenced by the number of validation windows: it degraded performance when there were two windows but had no impact with six. (iv) Raw features led to better test AP for LR compared to the main pipeline, although validation showed better results for the main pipeline. Such inconsistency highlights the importance of choosing a model based on validation data. These results guided the history-only policy, residual GCN, and top-k decision rule in Section 4.1.13.

Objective attainment. RO1 achieved (counts match; pipeline reproducible, including three-year extension); RO2 partially achieved (forecasting done and assessed, but no model did well and GCN failed to outperform the SVM); RO3 achieved (macro-F1 0.951 $>$ 0.70, though the threshold was shown to be weak); RO4 achieved (seeds, block bootstrap, permutations, rolling origin, spatial blocks, ablation studies, three-year replication); RO5 achieved in design and prototype of the batch job, API, and dashboard, but not in deployment.

Issues and challenges encountered. Table 4.22 shows the problems encountered during the project, how they were solved, and what is still unresolved.

Table 4.22: Issues and challenges encountered, responses and status

Issue	How it was detected	Response in the solution	Status
Circular Phase 1 label	Arg-max of pilot features reproduced the label for all 157 cells (E0)	History-only features; forecasting reframing; leakage rule documented	Resolved
Single random split, 5 positive test cells	Pilot recall moved 20 points per cell; LR matched pilot perfectly	Chronological 314-row test; rolling origin; block bootstrap	Resolved
Counts did not reconcile	228,635 expected vs 228,595 observed	Stage-wise log found 40 missing coordinates	Resolved
String-valued arrest/domestic counts	Saved pilot CSV had "NNYN…" values	Flags converted to 0/1 before aggregation	Resolved
Constant edge weights	All edges 2.2 km; weights unused by pilot	Weighted vs unweighted and larger graphs tested (E5)	Resolved (documented)
GCN over-smoothing	GCN below MLP; worse as degree grows	ResGCN with self path; graph ablations	Partly resolved (no graph gain)
Rare and singleton classes	One-cell SEX OFFENSE class; 27-row criminal-damage class	Training-only grouping rule; class weights; macro-F1	Partly resolved (weak minority F1)
Threshold instability	Tuned thresholds hurt test F1 with 2 validation periods, neutral with 6 (E4, E21)	Prevalence-matched top-k decision rule; tuning only with ≥ 6 validation periods	Resolved
Hyperparameter tension in proposal	Fixed paper settings vs promised search	Validation-only grid, 3 seeds	Resolved
Spatial dependence in evaluation	Neighbouring errors correlated; $n_{\text{eff}} \approx 47$–$50$ (E19)	Spatial-block bootstrap, permutation tests, block CV, three-year replication	Partly resolved (effective n stays small)
DBSCAN parameter sensitivity	PAI favoured tiny areas	Used as descriptive layer only	Unresolved for decisions
Reporting and enforcement bias	Recorded data reflect policing	Framing, governance and audit design (§4.4)	Unresolved (needs external data)
Single-year evidence	One-year extract, two test periods	Three-year replication and 12-window rolling origin (E21)	Resolved for Chicago; other cities untested
Preprocessing benefit asserted, not measured	Rubric check of Section 3.2.2	Ablation (E15): modest, model-dependent effect; claim narrowed	Resolved
Statements about the base paper and references	Re-reading the paper; online source check	Graph-statistics claim corrected; all 27 references checked against an external source	Resolved (all DOIs confirmed through Crossref or publisher pages)
Learned neighbourhoods and tuning as alternative explanations	Reviewer-style objections	GAT, centroid graph, tuned count models (E18–E19)	Resolved (graph still no gain)
Broad victimisation label	11 categories, 32% of records, very different offence types	Re-run with FBI index violent crime only (E22); ranking unchanged	Resolved
Unequal impact across areas	Flags concentrate where hardship is high	Error-disparity audit by hardship tercile (E23); low-reliability display for Phase 1; governance	Partly resolved (selection rate follows recorded base rates)
Limited model capacity as an explanation	Could a stronger learner fix Phase 1?	Gradient boosting tuned on validation (E24): no gain	Resolved (limit is data, not capacity)

4.3. Limitations

Data coverage and label construction. The main study uses one year of data from a rolling extract, which, according to CPD, is preliminary and not comparable over time; later reclassifications made after download are not reflected. The three-year replication includes two downloads spaced two months apart, and while there is no visible break in either counts or shares at the point where the two data sets are joined, minor differences in classification of data cannot be ruled out. The main test window covers a single summer (28 June to 22 August 2026), while the replication consists of six additional testing periods and a 12-period rolling origin that includes the summers of 2024–26. Victimisation rate uses the number of incidents committed against individuals and not victims, and the data has not been population or footfall adjusted, so locations with many people during the day, such as the Loop, would score highly because of visitor numbers rather than risk to residents. Homicides are counted per victim in the source, which is slightly inconsistent with the other categories.

Small effective sample and spatial autocorrelation. Although there are 228,595 events (732,017 in the replication), there are only 157–158 cells, and cell-periods are highly autocorrelated in time ($\rho \approx 0.96$) and space (Moran's $I \approx 0.5$). The heuristic effective sample size is around 47 for the test and 50 for the replication (E19), so additional periods do not contribute much new information. Dependence-aware intervals are wider than naïve intervals, and differences of roughly 0.01 AP are not resolvable. Independent information would require more cells, that is, more cities.

Temporal leakage and reproducibility. While the history-only constraint avoids feature leakage, the Phase 1 class categorisation and victimisation categories list were defined by the analyst with knowledge of the full category table (although only the training periods determined which categories were retained). The experiments E15–E19 and E22–E24 were planned after the main test outcomes were known; they employ validation data for all choices and do not alter the main findings, but are post-hoc and should be treated as such with respect to the test results. Repeating experiments E1, E3 and E10 with modified code yielded the same results as before while repeating experiments E4, E11 and E12 resulted in byte-identical outputs; neural nets were trained on CPU with the same seed every time, but are not necessarily bitwise repeatable across hardware.

Unmeasured confounding and features. Only crime-record features were used; features such as land use, population, places of interest, weather and events were not used; hence the model cannot differentiate between where the harm occurs and where it was reported. Feature importance is an indicator of predictive relationship and not causality.

Reporting bias, feedback loops and fairness. Recorded incidents reflect reporting practices and enforcement effort as well as actual incidents (Lum & Isaac, 2016; Richardson et al., 2019). Since the chosen model is based primarily on past counts, any implementation that in-

creases enforcement efforts to flagged cells may increase future counts in those cells and ensure that they remain in the top rankings (Ensign et al., 2018). No demographic variables were used as model inputs. An area-level error-disparity audit (E23, Section 4.1.15) found that Phase 2 recall and false-positive rates are close across hardship terciles, but that the model flags high-hardship areas about 2.7 times as often as low-hardship ones, following the recorded base rates. Because the census indicators are dated and the comparison is between areas rather than population groups, this is not a full group-fairness audit and no fairness claim beyond it is made. Area-level outputs reduce privacy risk relative to person-level prediction, but labelling areas as "high victimisation" may still cause stigmatisation of the population.

External validity. Results are specific to Chicago, 2023–26, a 2.2 km grid and 28-day periods. Transfer to New Zealand cities is untested.

4.4. Application Conversion

Figure 4.10 shows the proposed architecture of the application.

This section describes how the chosen solution would be converted into a working application. Nothing here has been deployed. The batch scoring job, read-only API and dashboard have been implemented as a prototype (Folder `Assessment2/app`) and validated automatically (Table 4.24); the rest are design only.

Users and workflow. The primary users are community safety or crime analysts who create a 28-day crime prevention plan like assigning outreach workers, conducting environmental design assessments, and victim visits. The process involves the following stages: (1) After each period, the model updates its data and ranks all the cells; (2) The analyst sees the top 32 cells ranked by the model with history and uncertainty; (3) The analyst logs her/his decision on whether to include, exclude, or review and why; and (4) After the end of the next period, results are analyzed relative to forecasts. The model never assigns any resources automatically.

Views. *Map view*: Cells colored based on forecast rank categories (Top 10%, 10-20%, etc.), along with the DBSCAN micro-hotspots overlay layer and the KDE surface overlay layer as optionals. *Ranking list*: Rank, cell identifier/name, score, change in rank (compared to last ranking period), and trend of last four victimisation periods. *Cell detail*: Sparkline of 13-period history, top three features as indicated by LR coefficients (Table 3.3), crime type forecast as "low reliability" (macro-F1 $\approx$ 0.5), and an evidence panel specifying model version, training periods, and testing performance. *Uncertainty*: rank intervals from bootstrapping the training cells, and a tag for cells within two ranks of the cut-off, since the 10 test cell-periods within two ranks of the cut-off (3% of rows) contain 4 of the 10 errors (40% error rate vs 2% elsewhere; E20, Figure 4.4A).

API. The read-only REST service exposes forecasts to the dashboard and other approved systems. The response below is illustrative (the values are not real forecasts).

```
GET /v1/forecasts?period=2026-08-23&top_k=32
200 OK
{ "period_start": "2026-08-23", "period_days": 28,
  "model": {"name": "lr-victimisation", "version": "1.0.0",
            "feature_schema": "fs-2026.1", "grid": "utm16n-2200m-v1",
            "trained_on_periods": "2025-11-16..2026-07-25"},
  "cells": [ {"grid_id": "12_7", "rank": 1, "score": 0.997,
              "band": "top10", "rank_interval": [1, 3],
              "history_victim_counts": [181, 177, 191],
              "near_cutoff": false,
              "top_factors": ["long-run victimisation mean",
                              "recent total incidents"]} ],
  "disclaimer": "Area-level recorded-incident forecast; not a
                 measure of individual risk." }
```

Input validation refuses periods that have not yet closed, unknown grid versions and requests for individual-level data. All responses are logged.

Preprocessing parity and versioning. The cleaning, projection, gridding, and feature code in this work are provided as one versioned library that is imported for both training and scoring, thus ensuring that the same set of transformations (boundary filter, EPSG:26916 projection, grid origin, log transform, scaler) are applied at inference time. The model registry contains the model, the fitted scaler, the feature schema hash, the grid and graph version, training periods, and test metrics all in one artifact; a scoring job refuses to run if any hashes differ.

Batch versus online inference. Forecasts get updated only upon the completion of a period, hence batch prediction for every 28 days, and also an optional "period-to-date" weekly prediction update are sufficient, so online inference is unnecessary. The measured cost (Table 4.12) is in milliseconds for city-wide prediction runs and under one second per retraining on a laptop processor. Thus the model can be run on a small virtual machine and an available analytics server. If no existing server is available, the smallest general-purpose cloud machine is more than enough: AWS's public price list (published 29 October 2026) gives on-demand Linux rates of US$0.0264 per hour for a `t3.small` (2 vCPU, 2 GiB) and US$0.0528 per hour for a `t3.medium` (4 GiB) in the Sydney region (Amazon Web Services, 2026). Table 4.23 turns these into monthly estimates. Staff effort, not computing, is therefore the main cost (Table 4.25).

Table 4.23: Running-cost estimate (assumptions: AWS Sydney on-demand Linux prices of 29 October 2026; 730 hours per month; storage, data transfer and tax excluded; prices change and should be re-checked before purchase)

Option	Hourly price (US$)	Hours per month	Monthly cost (US$)
`t3.small`, always on (API + dashboard + batch job)	0.0264	730	19.27
`t3.medium`, always on (headroom for retraining challengers)	0.0528	730	38.54
`t3.small`, batch job only (13 runs a year, 1 hour each)	0.0264	≈1.1	0.03

Graph models, new nodes and coverage changes. The ResGCN candidate uses a static and versioned grid graph. A cell that is newly created by an annexation or geocoding change will have no previous history, hence it gets a "cold start" flag, and the LR champion, which needs no neighbours, scores it using the available lags. The grid graph gets recreated, and the GCN gets trained only during the next scheduled release period. The change in the grid size or origin will cause a new graph version and requires full re-evaluation, because all features and labels change.

Monitoring, delayed labels and retraining. The labels are received 28 days after each forecast (plus the seven-day delay of the portal). For each closed period, the system calculates the AP, top-k precision, and recall with respect to the actual counts, and compares them with the acceptance bands based on the rolling origin analysis (AP ≥ 0.94, top-k precision ≥ 0.85). The feature drift is assessed via population stability indices for the main count features, and label drift via the percentage of cells switching bands. The model is retrained every third period or when an alert is triggered; the retrained challenger becomes the new champion if it equals or outperforms the current one in the last two closed periods. Every release keeps the previous version available for one-click rollback.

Security, access and audit. The access is limited to designated analysts via the single sign-on capability of the organization with role-based privileges (viewer, analyst, administrator). The application keeps only area-based aggregations, while the incidents are stored in the source system. All queries, exports, and analyst overrides are recorded in an append-only audit log, which is used in the bias assessment below.

Responsible-use safeguards. Interface labels describe outputs as forecasts of recorded incidents, neighbourhood names omitted from exports by default, and a required written reason for any action targeting a flagged neighbourhood. A quarterly check is done to see if forecasted bands align with independent metrics (e.g., hospital assault presentations or victimisation survey, when applicable) in order to see if there is any feedback loop (Ensign et al., 2018), and any enforcement-oriented use is paired with non-enforcement interventions. The use of a census-based group-fairness audit is a prerequisite for broader adoption. Human oversight is built into the design: forecasts are aggregate decision support only, a human analyst reviews every forecast and keeps the final decision, and no output is used to infer anything about an individual.

Built and tested prototype (E20). The prototype implements a batch job that fits the model and scores every cell for each period using only historical data. It also produces bootstrap rank intervals and explains the top factors behind the predictions. The prototype includes a read-only JSON API that follows the response schema above, API-key roles, input validation, an append-only audit log and an HTML dashboard with a ranked list and a map. At start-up, it rejects any prediction store whose feature-schema hash differs from the one expected by the code. The serving layer uses only the Python standard library. All 20 automated tests summarised in Table 4.24 passed. The parity test confirmed that the forecasts served for the two test periods

matched the offline E2 results exactly, with an AP of 0.983, a top-20% precision of 0.938 and a recall of 0.909. This agreement shows that the preprocessing used to serve forecasts matches the preprocessing used in training. The median latency for 600 mixed requests was 0.40 ms (95th percentile 0.50 ms) on the local machine, so it gives no information on network or load conditions; the prediction store size is 423 kB. The prototype has also demonstrated that the near-cut-off flag is a useful feature, since 10 of the 314 test cell-periods carry it and 4 of those 10 are misclassified, against 6 errors among the other 304. Not implemented: online inference, authentication beyond API keys, automated retraining, drift detectors, fairness review.

Table 4.24: Prototype tests (E20): 20 of 20 passed

Test group	Tests	What is checked
Access control	4	Missing and wrong keys rejected (401); viewer blocked from cell history (403); analyst allowed
Input validation	6	Unknown cell (404); non-integer, unknown period, $k = 0$, $k = 1000$ and unknown grid version (400)
Response contract	5	Health endpoint; unclosed period flagged; disclaimer, model and schema version present; City of Chicago data-modification notice present; ranks 1–32 without gaps
Serving parity	2	Served AP equals offline AP; served top-20% precision and recall equal offline
Safeguards	3	Dashboard renders with disclaimer; mismatched schema refused at start-up; every request in the audit log

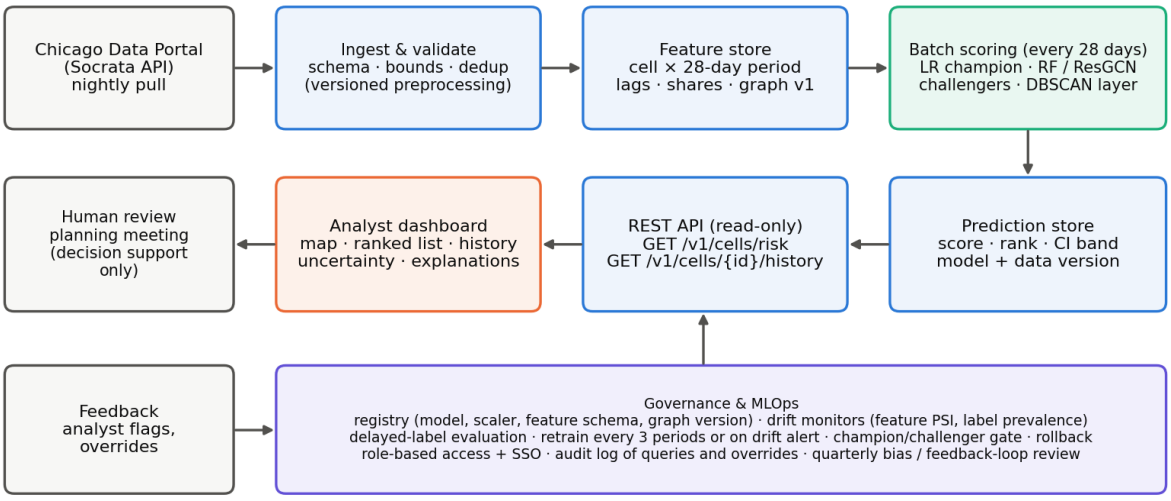


Figure 4.10: Proposed application architecture (design, not deployed).

Table 4.25: Staged implementation plan and feasibility (assumptions: one analyst-developer; existing server; open data)

Stage	Deliverable	Effort (estimate)	Acceptance criterion
0. Built here	Reproducible pipeline, models, evaluation, figures; prototype batch job, API and dashboard	Completed	Counts reconcile; test AP 0.983; 20 of 20 API tests pass; served metrics equal offline
1. Shadow mode	Nightly ingest, 28-day batch scoring, internal dashboard; no operational use	3–4 weeks	3 consecutive periods with AP ≥ 0.94 and no pipeline failures
2. Pilot with analysts	Ranked list, cell detail, override logging, monitoring	4–6 weeks	Analysts rate usefulness; overrides reviewed; no drift alerts unexplained
3. Governed release	SSO/RBAC, registry, rollback, quarterly bias review	4 weeks	Governance sign-off; E23 disparity audit repeated on current census data
4. Extensions	Covariates (land use, population), multi-year data, ResGCN challenger	Ongoing	Challenger beats champion on two closed periods

Innovation and problem-solving. These design decisions solve the problems raised in this study. Top-k solves the problem of instability of the threshold as seen in E4, whereas near-cutoff takes care of the problem depicted in Figure 4.4. Cold start solves the problem of GCN dependency on fixed graph, while preprocessing parity solves the problem of pilot's flaws at file level. Delayed label checking and champion-challenger gating let more sophisticated models become eligible when they bring in value.

4.5. Chapter Summary

Predictions evaluated in chronological order using a leakage-free procedure indicate that observed victimisation hot spots are very predictable (AP 0.983, macro-F1 0.951), and persistence explains much of the predictive power. The pilot version of the GCN failed due to over-smoothing of the cell-level signal by symmetric normalisation. The residual GCN solved the issue, yet the evaluation revealed no gains from considering neighbouring cells. It is difficult to forecast the dominant crime type. The results are replicated for the narrower category of violent crimes and for the gradient-boosting learner. The area-level disparity audit reveals similar error rates but different selection rates across hardship terciles. The chosen logistic regression model is fast to compute and interpretable. The application design incorporates versioning, monitoring and governance to support responsible use of the model as a decision-support tool.

Chapter 5. Conclusion

The aim of this project was to forecast where recorded crime and recorded victimisation will concentrate in Chicago using 2.2 km cells so as to inform prevention planning. In Assessment 1, spatial GCNs were used to solve two tasks, and encouraging results from the pilot experiment were presented. In this study, the audit of the pilot showed that the Phase 1 task was circular; that the Phase 2 evaluation used only five positive test cells; that LR and persistence were able to match or beat the pilot GCN result. A leakage-safe forecasting pipeline was then developed: 228,595 reconciled events, 157 cells, thirteen 28-day periods, 23 history-only features, ten model families with the same tuning budget, chronological testing, ten random seeds, and spatial-block uncertainty estimates.

The confirmed results are clear. Cells that will fall into the top 20% for recorded victimisation in the next period are predicted with high precision (LR: AP 0.983, macro-F1 0.951, precision 0.938, recall 0.909), much higher than the pre-specified benchmark, but a recency-weighted persistence method is almost equally accurate (AP 0.973), and the permutation tests find no significant differences between LR, RF, MLP, SVM, residual GCN, and persistence. A three-year replication using six test windows yielded similar results (RF 0.965, residual GCN 0.964, LR 0.958, persistence 0.951; GCN 0.875, GAT 0.870), and in a 12 window rolling origin LR outperformed persistence only 6 times. The pilot-style GCN under-performed (AP 0.885) because symmetric normalisation on the rook lattice gives each cell only about a fifth of the weight on its own history; graph ablations confirmed that performance falls as neighbourhoods widen or layers are added. The residual GCN, derived from this analysis, recovers the loss (AP 0.975) but does not benefit from neighbouring cells, and neither does a graph attention network, a weighted centroid graph, hand-crafted spatial-lag features or spatial-block transfer. The dominant crime type in the next period is much harder to predict (highest macro F1 about 0.50). Improvements in LR over persistence are suggestive but not significant (permutation $p = 0.077$) while additional features helped RF but not the neural networks. Clusters generated by DBSCAN (unsupervised) are useful for describing micro-hotspots but less effective than ranked cells for area planning (PAI 1.84 vs 2.59). The same ranking of methods held with a much narrower violent-crime label (E22), gradient boosting added no skill (E24), and an area-level audit found similar Phase 2 error rates across hardship terciles but flags concentrated where recorded victimisation and hardship are highest (E23).

The chosen solution is thus a transparent, regularised logistic regression ranking model with persistence and KDE as reference layers, RF and ResGCN as monitored challengers, and a governed application that provides ranked, uncertain, area-level forecasts to analysts. A prototype for the batch job, API, and dashboard was constructed and tested (20 out of 20 tests; forecasts served by the prototype match the offline metrics exactly). Objectives RO1, RO3, and RO4 were reached, RO2 was partly reached, and RO5 was reached as a design with a tested prototype. These problems included label leakage, a small pilot test set, irreconcilable counts, malformed flag features, constant graph weights, over-smoothing, unstable decision thresholds and some

overly-strong claims in the literature review in the proposal—and were all identified through explicit checks and mitigated (Table 4.22). The reporting bias and feedback loop problem are still open, and have governance solutions only.

The study provides public-sector analysts with an inexpensive, interpretable ranking validated on held-out data. It also offers evidence about when a more complex model may offer too little benefit to justify its cost. The findings support short-horizon, area-level forecasting of *recorded* incidents in Chicago using data from 2023–26. Performance in other cities, at finer spatial scales and in operational use remains untested. In practice, actions taken in response to forecasts may change the data recorded later.

Future work. Each item follows from a specific finding:

1. **More cities and longer histories.** The results prove to be consistent for the three-year replication process in Chicago, but the differences of about 0.01 AP still cannot be resolved since the number of cells, not periods, restricts the evidence (E19). Other cities would add new independent cells and conduct transfer testing.
2. **Non-crime covariates**, such as (population, land use, public transport, and points of interest), as per Zhang et al. (2020), are used to distinguish where crime happens and where crime was reported, and to compute population-adjusted rates.
3. **Graph models that learn the neighbourhood** attention-based or learned-adjacency GNNs, and spatio-temporal models like the one by Han et al. (2020) validated against the very strong persistence and LR baselines we have set here.
4. **Count-based ranking** with properly tuned Poisson or negative-binomial models, because counts provide more information than just being in the top 20%.
5. **A fuller fairness and feedback audit** that extends the area-level check of E23 with current census-tract data and independent victimisation measures (surveys, hospital data) before any operational use.
6. **A shadow-mode deployment** (Stage 1 of Table 4.25) to measure performance on genuinely future data and analysts' use of the outputs.

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Chapter A. Data Dictionary

Table A.1: Model features (computed per cell for target period t from periods $< t$ only)

Feature	Definition	Transform
VICTIM_L1, _L2, _L3	Victimisation-related incidents in periods $t-1$, $t-2$, $t-3$	$\log(1+x)$, z-score
TOTAL_L1, _L2, _L3	All incidents in periods $t-1$, $t-2$, $t-3$	$\log(1+x)$, z-score
VICTIM_MEAN_ALL	Mean victimisation incidents per period over periods $0, \ldots, t-1$	$\log(1+x)$, z-score
TOTAL_MEAN_ALL	Mean incidents per period over periods $0, \ldots, t-1$	$\log(1+x)$, z-score
VICTIM_SHARE	Victimisation share of all incidents in periods $< t$	z-score
DOMESTIC_SHARE	Share flagged domestic in periods $< t$	z-score
ARREST_SHARE	Share with arrest in periods $< t$	z-score
NIGHT_SHARE	Share occurring 20:00–05:59 in periods $< t$	z-score
VICTIM_TREND	VICTIM_L1 − VICTIM_L3	signed log, z-score
SHARE3_{category} (8)	Share of incidents in periods $t-3..t-1$ that are theft, battery, criminal damage, assault, motor vehicle theft, other offence, deceptive practice, burglary	z-score
LAT, LON	Cell-centre coordinates (WGS84)	z-score
SLAG_VICTIM_L1, SLAG_VICTIM_MEAN	Row-normalised neighbour average of VICTIM_L1 / VICTIM_MEAN_ALL (E5 ablation only)	$\log(1+x)$, z-score
Y_VICTIM	Target: victimisation incidents in period t	–
HIGH	Target: Y_VICTIM $\geq$ 80th percentile of the 157 cells in period t	–
Y_P1	Target: dominant category in period t mapped to battery, criminal damage, theft, other	–

Table A.2: Raw schema of the one-year CPD extract (17 fields, 229,275 rows) and how each field was used

Field	Missing (raw)	Use
CASE#	0	Duplicate check only (22 repeated values kept: one case can have several records)
DATE OF OCCURRENCE	0	Parsed to a timestamp; defines the 28-day periods and the night-time share
BLOCK, IUCR, FBI CD	0	Not used (block-level address and classification codes)
PRIMARY DESCRIPTION	0	Victimisation flag; crime category (categories under 1,000 incidents grouped as OTHER); Phase 1 label
SECONDARY DESCRIPTION	0	Not used
LOCATION DESCRIPTION	1,037	Set to UNKNOWN; not used in the final features
ARREST, DOMESTIC	0	Converted to 0/1; arrest and domestic shares
BEAT, WARD	0 / 1	Not used (administrative areas)
X / Y COORDINATE, LOCATION	40	Not used (the 40 rows also lack latitude and longitude and were removed)
LATITUDE, LONGITUDE	40	Projected to EPSG:26916; grid and graph construction; cell-centre coordinates

Why these victimisation categories. The 11 types are the main categories of crimes where the victim is a person (violence including assault and battery, robbery, sexual offences, stalking, intimidation, kidnapping, human trafficking, offenses involving children); property, drug, weapons possession, and public-order offenses are not included, as they often lack any identifiable victim. This classification is rather general since SEX OFFENSE and OFFENSE INVOLVING CHILDREN include quite distinct crimes, and 32% of all records fit into this category. A narrower

definition using FBI index violent crime only was tested in E22 (Section 4.1.15) and gave the same ranking of methods.

Victimisation categories: assault, battery, robbery, criminal sexual assault, sex offence, stalking, homicide, kidnapping, offence involving children, intimidation, human trafficking. Raw fields used: `DATE OF OCCURRENCE`, `PRIMARY DESCRIPTION`, `LOCATION DESCRIPTION`, `ARREST`, `DOMESTIC`, `LATITUDE`, `LONGITUDE`.

Chapter B. Split Manifest and Configurations

Table B.1: Period manifest (28-day periods from 24 August 2025)

Period	First day	Last day	Incidents	Role
0	2025-08-24	2025-09-20	18,683	features only
1	2025-09-21	2025-10-18	18,960	features only
2	2025-10-19	2025-11-15	18,236	features only
3	2025-11-16	2025-12-13	16,157	train
4	2025-12-14	2026-01-10	16,164	train
5	2026-01-11	2026-02-07	15,057	train
6	2026-02-08	2026-03-07	16,488	train
7	2026-03-08	2026-04-04	17,576	train
8	2026-04-05	2026-05-02	17,722	train
9	2026-05-03	2026-05-30	18,970	validation
10	2026-05-31	2026-06-27	18,399	validation
11	2026-06-28	2026-07-25	18,938	test
12	2026-07-26	2026-08-22	17,223	test
(13)	2026-08-23	2026-08-23	22	not modelled

Three-year manifest. Periods are 28 days from 24 August 2023 (period 0); features-only periods 0–2; training target periods 3–26 (16 November 2023 to 17 September 2025); validation 27–32 (18 September 2025 to 4 March 2026); test 33–38 (5 March to 19 August 2026); 4 leftover days (22 August onwards) are not modelled. Cells: 158; undirected edges: 274.

Fixed configuration. Grid 2,200 m, origin (425,477; 4,610,479) in EPSG:26916; graph radius 3,000 m (ablations 3,200; 4,500; 6,600 m); Phase 1 rare-category threshold 1,000 incidents and class-keeping threshold 5% of training cell-periods; Phase 2 label percentile 80 (ablations 70, 90); neural training: Adam, at most 400 epochs, patience 60, full batch, class-weighted cross-entropy; seeds 0–9 (tuning 0–2); bootstrap 2,000 resamples of 3×3-cell blocks; DBSCAN grid $\varepsilon \in \{100, 150, 200, 300\}$ m $\times$ minPts $\in \{30, 60, 100\}$. Selected hyperparameters are listed in Table 3.4; full tuning results are in `results/E1_p2_tuning.csv` and `results/E3_p1_tuning.csv`.

Table B.2: Sensitivity analysis (E7): Phase 2 test AP / ROC-AUC

Setting	Persistence	LR	GCN (5 seeds)	ResGCN (5 seeds)
70th percentile (prevalence 0.30)	0.980 / 0.991	0.983 / 0.992	0.921 / 0.965	0.978 / 0.991
80th percentile (prevalence 0.21)	0.973 / 0.993	0.983 / 0.996	0.896 / 0.962	0.975 / 0.993
90th percentile (prevalence 0.10)	0.930 / 0.991	0.906 / 0.985	0.809 / 0.968	0.886 / 0.981
1.5 km cells (312 cells)	0.956 / 0.986	0.963 / 0.989	0.841 / 0.948	0.957 / 0.987
2.2 km cells (157 cells)	0.973 / 0.993	0.983 / 0.996	0.896 / 0.962	0.975 / 0.993
3.0 km cells (95 cells)	0.968 / 0.992	0.965 / 0.991	0.843 / 0.955	0.968 / 0.992

Chapter C. Code Structure and Reproduction

The code, results, figures and this report are available at `https://github.com/saminatul/IT8105-20251674-samina-area-level-crime-hotspot-prediction`.

```
Assessment2/
  code/prep.py              cleaning, projection, grid, graph, features
  code/models.py            metrics, baselines, KDE, GCN, ResGCN, MLP, training
  code/run_experiments.py   E0-E19, E21 (A2_DATA=multi for 3 yrs)
  code/extensions.py        E22-E24 (violent-only label, disparity audit, boosting)
  code/figures.py           all figures in figures/
  results/                  CSV/JSON outputs and seed-level predictions
  code/multiyear.py         three-year data loader and split
  app/                      score_batch.py, server.py, test_app.py (prototype)
  results_multi/            outputs of the three-year runs
  figures/                  PNG figures used in this report
  latex/                    main.tex, references.bib, figures/
machinelearningcode/Coding/Dataset/
  Crimes_-_One_year_prior_to_present_20260831.csv  (input)
  external/census_socioeconomic_2008_2012.csv      (E23)
  external/community_areas.geojson                 (E23)
```

Run instructions. Install Python 3.9+ with `numpy`, `pandas`, `scikit-learn`, `torch`, `torch-geometric`, `pyproj`, `imbalanced-learn`, and `matplotlib`, then execute `python3 run_experiments.py` and `python3 figures.py` in `Assessment2/code`. The pipeline starts from the initial CSV, recreates all intermediate tables in memory, and outputs all results used in this report. The seeds and splits are fixed so that re-runs will produce identical tables, with differences only due to floating-point inaccuracies. For the three-year replication run `A2_DATA=multi python3 run_experiments.py E1 E3 E4 E8 E9 E10 E16 E19` (and `A2_ROLL=15:39:2 A2_SUFFIX=_long` with `E8` for the long rolling origin); this needs the second CSV described in Section 3.2. Extension experiments can be done using the command `python3 extensions.py` after the main experiment (E23 requires two City of Chicago datasets in `Dataset/external`). The prototype is built with `python3 score_batch.py` and tested with `python3 test_app.py` from `Assessment2/app`. No credentials or personal identifiers are stored; test API keys are generated at run time and never saved; the outputs are cell-level aggregates. The terminal transcripts saved from the runs performed for this submission are in Appendix G.

Chapter D. Learning Outcomes and Requirements

Table D.1: Learning-outcome coverage

Learning outcome	Where addressed	Evidence
LO1: Analyse issues and challenges of ML algorithms for optimising applied industry solutions	§4.1.2, §4.1.6, Table 4.22, §4.3, §4.4	Leakage, small-sample, over-smoothing, threshold-instability and bias issues detected with evidence and resolved or bounded (including the area-level error-disparity audit, E23, and label-definition check, E22); deployment issues (drift, delayed labels, new nodes) designed for
LO2: Evaluate mathematical and statistical background to optimise solutions	§3 Eqs. 3.1–3.8, §4.1.8, Table 4.10	Self-weight analysis of GCN normalisation led to ResGCN and graph ablations; KDE bandwidth bias–variance; regularisation C and weight decay; metric choice under imbalance; block bootstrap and paired differences
LO3: Apply design solutions using supervised and unsupervised algorithms	§4.1.3, §4.1.4, E12, §4.4	Supervised: LR, SVM, RF, HGB, MLP, GCN, ResGCN, GAT, Poisson GLM on two tasks (E2, E3, E24). Unsupervised: DBSCAN clustering (evaluated by PAI and stability) and KDE density estimation (evaluated on the same task as supervised models; whether KDE alone counts as unsupervised under the lecturer's interpretation is uncertain and flagged for confirmation). GCN training is supervised and is not counted as unsupervised

Table D.2: Assessment requirements mapped to report sections and evidence

Requirement	Section	Evidence
Solution designed and applied (more than two algorithms)	§4.1.3–4.1.4	Ten model families (including gradient boosting, E24) plus DBSCAN and count regression, all implemented and tested
Methodology described and justified	Chapter 3	Pipeline, objective mapping, protocol, grids, equations
Data preprocessing	§3.2	Reconciled counts, projection, grid, graph, features, imbalance, importance
Wide range of experiments	Table 4.1	25 experiment groups (E0–E24), seeds, ablations, sensitivity, robustness, label-definition and disparity checks
Statistical evaluation for optimisation	§4.1.8, Table 4.10	Block bootstrap CIs, paired differences, validation-only tuning
Critical evaluation of algorithms	§4.1.6, Discussion	Strengths, failure modes, cost (Table 4.12)
Issues and challenges	Table 4.22, §4.3	Detected, responded, status
Conversion into usable application	§4.4	Workflow, views, API, versioning, monitoring, security, plan; prototype tested (Table 4.24)

Chapter E. Traceability to Assessment 1

Table E.1: Assessment 1 claims and plans checked against the final study

Assessment 1 statement or plan	Where treated	Outcome
Pilot data: 229,275 raw, 228,595 cleaned records, 24 Aug 2025–23 Aug 2026	Table 3.2	Confirmed; the proposal omitted 40 records without coordinates
157 cells, 544 directed / 272 undirected edges, no isolated nodes	§3.2.2	Confirmed; every edge is 2.2 km, so the weights are constant
Phase 1: 23 features, 7 classes, 110/23/24 split, GCN 66.67% vs SVM 37.50%, macro-F1 0.218	Table 4.2	Reproduced as circular (label = arg-max of input features); replaced by 4-class forecasting
Phase 2: 10 features, 109/24/24 split, GCN accuracy 0.875, macro-F1 0.842, AUC 0.989, PR-AUC 0.967	Table 4.2	Not re-run; LR and persistence score 1.000 on the same split; replaced by chronological testing
KDE accuracy 94.9% on the binary hotspot task	§4.1.3	Not comparable with multiclass scores; KDE re-evaluated on the same tasks as other models
Bootstrap on one 24-cell test set rules out a lucky split	§4.1.2	Claim withdrawn; replaced by block bootstrap, permutation tests, seeds and replication
RO1–RO5, RQ1–RQ3	Table 1.1, §4.2	Retained, made measurable; statuses reported
Planned: SMOTE vs class weights	§3.2.3, E4	Done on LR only (no valid synthetic edges for GCN)
Planned: stratified 5-fold CV, LOOCV, nested CV	§3.5	Not used: temporal leakage, singleton classes, validation-then-test already separates tuning
Planned: time-series CV	E8, E21	Done as rolling origin (5 windows; 12 windows on three years)
Planned: grid lr $\{0.001, 0.01\}$, hidden $\{64, 128\}$, dropout $\{0.3, 0.5\}$	§3.5	Superset used, except lr 0.001 was not searched
Planned: SHAP	§3.2.6, E14	Replaced by coefficients and permutation importance
Planned: MAE, RMSE, R^2 for a continuous score	E11, E18	Done (tuned follow-up)
Planned: fairness audit	§4.1.15, §4.3	Area-level error-disparity audit done (E23); full group-fairness audit still needs current demographic data; no fairness claim beyond E23
Cited *Crimes – 2001 to Present* as the data source	§3.2	Corrected: one-year extract used (plus the 2001-to-present subset for E21)
Lum and Isaac (2016) "analyzing Chicago" and the Strategic Subject List	§2.2	Corrected: they analysed Oakland drug-crime data; the Chicago statements are re-attributed to Richardson et al. (2019)
Base-paper graph: 487 nodes, 5,832 edges, average degree 11.97	§2.2	Kept; clarified that edges are counted as directed pairs and that centroid-based distances matter
Learning outcomes: LO1 and LO2 only	App. D	Extended to LO1–LO3
Limitations: one year, 157 cells	§4.1.14	Three-year replication added; cell count remains the limit

Chapter F. Evidence Inventory and Unresolved Gaps

Table F.1: Evidence inventory

Status	Items
Available and used	One-year CPD extract; 2023–25 subset of the 2001-to-present dataset; all code (`code/`, `app/`); result files for E0–E24; City of Chicago census indicators (2008–12) and community-area boundaries (E23); 15 figures; Assessment 1 proposal and notebooks; four supplied papers; terminal transcripts of the runs performed for this submission (Appendix G)
Reproduced from Assessment 1	Cleaning counts, graph size, pilot split and Phase 2 LR result (Table 4.2); the pilot GCN numbers themselves were not re-run
Not independently verified	Lecturer's name (taken from the proposal; confirm before submitting); Silverman (1986) publisher details (no DOI exists; standard citation used)
Missing	Lecturer feedback; population and land-use data; current tract-level demographic data; data from other cities; a deployed pilot

Table F.2: Unresolved evidence gaps and the work needed to close them

Gap	Work needed	Affects
No model reliably beats persistence for victimisation	More independent cells (other cities) or covariates; larger test of the 0.01 AP increment	Solution, Experiments
Phase 1 crime-type forecasting is weak and unstable	Population and land-use covariates; spatio-temporal models; more cells	Solution
Only about 47–50 effective observations	Data from additional cities	Statistics
Recorded incidents reflect reporting and enforcement	External victimisation measures (surveys, hospital data)	Issues, Application
Only an area-level error-disparity audit (E23)	Current census-tract data and a group-fairness audit protocol	Issues
Prototype not deployed; no priced cost estimate	Shadow-mode pilot with an agency; verified cloud prices	Application

Chapter G. Terminal Transcripts of the Runs Performed

The transcript files below contain terminal output saved while the experiments were run from `Assessment2/code`. Each command is the one that wrote its file. Available transcripts are included unchanged; if a file is not bundled, its absence is noted. These logs cover the post-hoc experiments, the three-year replication, the 12-window rolling origin, and the extension experiments. The result tables in the report are the CSV and JSON files those runs wrote, not a retyping of the screen.

Post-hoc experiments (E14–E19).

```
python3 -u run_experiments.py E14 E15 E16 E17 E18 E19 2>&1 | tee ../evidence/G_posthoc.txt
```

Output saved as `evidence/G_posthoc.txt`: *Transcript file not bundled with this document.*

Three-year replication.

```
A2_DATA=multi python3 -u run_experiments.py E1 E3 E4 E8 E9 E10 E16 E19 E13 2>&1 | tee
↪  ../evidence/H_multiyear.txt
```

Output saved as `evidence/H_multiyear.txt`: *Transcript file not bundled with this document.*

Twelve-window rolling origin.

```
A2_DATA=multi A2_ROLL=15:39:2 A2_SUFFIX=_long python3 -u run_experiments.py E8 2>&1 | tee
↪  ../evidence/H_rolling_long.txt
```

Output saved as `evidence/H_rolling_long.txt`: *Transcript file not bundled with this document.*

Extension experiments (E22–E24).

```
python3 -u extensions.py 2>&1 | tee ../evidence/I_extensions.txt
```

Output saved as `evidence/I_extensions.txt`: *Transcript file not bundled with this document.*